

Project Title

Detection of Autism in Toddlers Using Machine Learning.

By

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Dissertation Submitted to the University of Derby in Partial Fulfilment of the Requirements for the Award of Master of Science in Big Data Analytics

2023-2024

Abstract

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental disorder where early detection is crucial for implementing timely interventions that can significantly improve developmental outcomes in toddlers. Traditional diagnostic methods, while valuable, often fall short in terms of precision and scalability in varied clinical environments. To address these limitations, this research explores machine learning (ML) algorithms to enhance both the accuracy and interpretability of ASD detection methods. The study employs the Quantitative Checklist for Autism in Toddlers (Q-CHAT) dataset to evaluate the performance of several ML models, including Support Vector Machines (SVM), Random Forest, Logistic Regression, Decision Trees, and ensemble approaches. Local Interpretable Model-agnostic Explanations (LIME) were used to increase the model transparency and make complex predictions more understandable for clinicians. Key predictors of ASD were identified through advanced methods such as Recursive Feature Elimination (RFE), K-mean Clustering, and Principal Component Analysis (PCA), aiding in personalized intervention strategies. The study implemented 5-fold cross-validation to ensure the robustness and generalizability of the models, reducing the impact of overfitting and providing a reliable assessment of model performance across different data subsets. SMOTE was used to resolve the imbalance target variable. The findings demonstrate that the SVM model excelled, achieving 99.2% accuracy, 98.4% precision, and 100% recall, establishing it as the most effective classifier for ASD detection. Furthermore, a Stacking Classifier that integrated multiple models with a Logistic Regression meta-classifier exhibited outstanding performance, with a test accuracy of 98.9% and a perfect ROC AUC score of 1.000. These outcomes underscore the potential of ML models to transform ASD diagnostics by providing a more objective, scalable, and precise result. This research enhances ASD detection in toddlers and bridges the gap between advanced computational methods and practical clinical applications in medical informatics. The study highlights the need to improve model interpretability and advocates for the ethical use of AI in healthcare to ensure these tools are accessible and effective in clinical settings.

Keywords: *Autism Spectrum Disorder (ASD), Machine Learning, Support Vector Machine (SVM), Q-CHAT (Quantitative Checklist for Autism in Toddlers), Model Interpretability (LIME)*

Acknowledgment

I want to express my deepest gratitude to all those who have supported and guided me throughout the completion of this dissertation.

First and foremost, I thank the Lord God Almighty for granting me the wisdom, knowledge, and strength to complete this work. I am profoundly grateful to my wife, Elizabeth, whose love, patience, and encouragement have been my greatest strength throughout this journey.

I also extend my heartfelt appreciation to my supervisor, Dr. Mohsen Farid, for his expert guidance, insightful feedback, and unwavering support, which were instrumental in completing this research. I thank the University of Derby for providing the essential resources, facilities, and academic environment necessary for my research.

Additionally, I would like to thank my friends and colleagues for their continuous support and encouragement, which has been invaluable during this process. To all who have contributed to this project, your support has been crucial to its success, and I am deeply grateful.

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1. Chapter One: Introduction

Autism Spectrum Disorder (ASD) is a widespread neurodevelopmental disorder characterized by difficulties in social communication, restricted interests, repetitive activities, and a variety of strengths that usually develop in early infancy (Grabrucker, 2021). The Diagnostic and Statistical Manual of Mental Disorders, 5th edition (DSM-5), classifies ASD as a group of four disorders that share common challenges in communication, social interaction, and behavior. The term "spectrum" highlights the wide variety of symptom presentation and intensity observed in people with ASD diagnosis. Since its discovery in the mid-20th century, Autism Spectrum Disorder (ASD) has gained popularity because of its prevalence and complicated impact. Although the exact cause of Autism Spectrum Disorder (ASD) is still unknown, research has indicated that it most likely results from a combination of environmental and genetic factors. An individual's risk of developing ASD is increased by certain gene alterations, such as those linked to tuberous sclerosis and fragile X syndrome (Budha Gowtham's Lab et al., 2024). A person's susceptibility to ASD is also influenced by their living circumstances, family history, issues at birth, and whether their parents had children later in life. It's crucial to remember that scientific studies have not discovered any link between autism spectrum disorders and immunizations.

Despite continuous studies into the intricate interaction of genes and environment in autism, the number of cases globally continues to rise. Autism has increased by 178% during the last two decades. Autism affects one in every hundred children under the age of ten in India. A mix of genetic and environmental factors likely causes ASD. While over 420 genes have been related to autism spectrum disorder, specific mutations in 100 genes and 30 disease-causing genes are closely connected with neuronal development and functioning. These genetic alterations are responsible for around 70-80% of ASD cases. Environmental impacts are another element that contributes, making up 20–30% of the total. the mother's nutritional state, the father's age at conception, thyroid problems or gestational diabetes, Obesity in mothers, depriving the newborn of oxygen before, during, or following birth, and little time between pregnancies—less than a year are also factors that contribute to ASD cases (Budha Gowtham's Lab et al., 2024).

1.1 Project Rationale

The detection of Autism Spectrum Disorder (ASD) in toddlers (ages 18 – 36 months) presents a critical challenge in developmental healthcare, as early diagnosis is pivotal for initiating interventions that significantly enhance long-term outcomes in cognitive, social, and communicative domains. The Quantitative Checklist for Autism in Toddlers (Q-CHAT) is a widely recognized screening tool designed to quantify behaviors and symptoms associated with ASD in young children. However, traditional methods of analyzing Q-CHAT data are often labor-intensive and subject to human biases, potentially delaying the early detection and treatment that are vital for affected children.

The advent of machine learning (ML) offers the transformative potential to revolutionize the early detection of ASD by providing an objective, efficient, and scalable analytical framework. This project seeks to leverage advanced ML techniques to analyze the Q-CHAT dataset, aiming to develop a robust predictive model that enhances the accuracy and speed of ASD diagnosis. The integration of ML into this domain addresses several pressing issues: reducing the subjectivity inherent in clinical assessments, expediting the diagnostic process, and ensuring a more standardized approach to early ASD detection.

1.2 Objectives of the Study

The main objectives of the study are to:

- **Data Preprocessing and Cleaning:** Methodically clean and preprocess the Q-CHAT dataset to address any inconsistencies, handle missing values, and ensure the data is in a format suitable for ML analysis.
- **Feature Engineering and Selection:** Use feature selection approaches to find the best predictive variables, then conduct in-depth feature engineering to extract meaningful features from the Q-CHAT responses.
- **Model Development and Training:** Develop and train various machine learning models, such as logistic regression, k-nearest neighbors, decision trees, gradient boosting, random forests, support vector machines, and stacking classifiers. Each model will undergo thorough testing and validation to determine its prediction performance.

- **Model Evaluation and Validation:** Evaluate the models using reliable measures including accuracy, precision, recall, F1-score, and ROC-AUC. Cross-validate the model to verify it is generalizable and reliable across diverse subsets of data.
- **Interpretability and Explainability:** Implement LIME (Local Interpretable Model-agnostic Explanations) to provide interpretability to the ML models, ensuring that the decision-making process is transparent and actionable for healthcare professionals.

1.3 Scope of Exploration

Using the Q-CHAT (Quantitative Checklist for Autism in Toddlers) dataset obtained from (Niedźwiecka and Pisula, 2022), this project comprehensively explores a wide range of machine learning algorithms and explanatory methodologies to improve the early detection of Autism Spectrum Disorder (ASD) in toddlers. A simple method for binary classification is provided using logistic regression, which predicts the risk of ASD based on Q-CHAT responses and predictor variables (Tartarisco et al., 2021). To handle complex interactions in the dataset, a Support Vector Machine (SVM) is utilized to find the best hyperplanes dividing ASD risk groups (Alsuliman and Al-Baity, 2022). Random Forest improves predictive accuracy by combining predictions from many decision trees, whereas Decision Trees provide a simple and understandable structure for predicting ASD risk (Hemu et al., 2022).

The study also uses Gradient Boosting Machines (GBMs), to analyze the Q-CHAT dataset to enhance early Autism Spectrum Disorder (ASD) detection. GBMs offer high accuracy and adaptability, making them ideal for improving predictive performance in this project (Natekin and Knoll, 2013). K-Nearest Neighbors (KNN) classifies ASD risk by comparing samples to their closest neighbors, providing an intuitive, non-parametric prediction method. Additionally, K-Means Clustering is used to uncover distinct groups within the dataset, helping to identify varying levels of ASD risk and underlying data patterns. Rasch Analysis is used to assess the reliability and validity of Q-CHAT responses, enhancing the accuracy of ASD detection. Principal Component Analysis (PCA) is used to minimize dataset dimensionality by converting features into orthogonal components, collecting the most significant variation, and simplifying the dataset for better model performance (Bharadiya, 2023).

Furthermore, ensemble methods are used to integrate several algorithms, improving prediction performance by combining alternative approaches to data analysis (Twala and Molloy, 2023).

LIME (Local Interpretable Model-agnostic Explanations) is used to make complex model predictions more understandable by training smaller models on locally disturbed data (Pedersen and Benesty, 2018).

This study's integrated approaches attempt to advance early ASD identification by applying various machine-learning techniques. The goal is to improve diagnostic accuracy, accelerate the identification process, and support clinical decision-making using interpretable and robust models.

1.4 Significance of Study

This study aims to revolutionize early Autism Spectrum Disorder (ASD) detection in toddlers using advanced machine learning techniques on the Q-CHAT dataset by:

- **Enhancing Diagnostic Precision:** The study aims to improve the accuracy of ASD detection in toddlers by utilizing a variety of complex machine learning algorithms, such as SVM, K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting Machines, Logistic Regression, Stacking Classifier and K-mean Clustering. This will enable more accurate and dependable diagnoses.
- **Enhancing Interpretability of Complex Machine Learning Models:** Using LIME approach makes models more transparent. By offering concise, useful information that helps medical practitioners comprehend and believe in model projections, this increase in transparency promotes clinical adoption.
- **Uncovering Critical Predictors:** Using techniques such as Principal Component Analysis (PCA) and other sophisticated analytical approaches, the study identifies critical factors in the Q-CHAT dataset that have a substantial influence on ASD outcomes. This research informs personalized intervention strategies and aids in the formulation of successful policies for early ASD detection and treatment.

1.5 Contribution to Knowledge

This research makes significant contributions to Autism Spectrum Disorder (ASD) detection in toddlers by leveraging machine learning techniques to improve diagnostic accuracy, speed, and interpretability. Several algorithms, including logistic regression, decision trees, gradient-boosting machines, random forests, support vector machines, k-nearest neighbors, and stacking, were

applied to enhance the precision of early ASD detection. The study also employed LIME to increase model transparency and make complex predictions more understandable for clinicians. Key predictors of ASD were identified through advanced methods such as feature selection and Principal Component Analysis (PCA), aiding in the development of personalized intervention strategies. The study implemented cross-validation to ensure the robustness and generalizability of the models, reducing the impact of overfitting and providing a reliable assessment of model performance across different data subsets. By minimizing subjectivity and bias through objective machine learning analysis, the research contributes to a more standardized and efficient diagnostic process. Overall, this study provides valuable insights for improving early ASD detection in toddlers and supports the creation of more effective, individualized treatment approaches.

1.6 Organization of Thesis

1.6.1 Chapter 1: Introduction

This chapter introduces Autism Spectrum Disorder (ASD) and the need for improved early diagnosis. It explains the rationale for using machine learning to enhance the analysis of the Q-CHAT dataset. Key objectives include data preparation, model development, and evaluation. The chapter concludes by highlighting the study's potential to improve diagnostic accuracy and efficiency.

1.6.2 Chapter 2: Literature Review

This chapter reviews research on Autism Spectrum Disorder (ASD), including its history, diagnosis, and prevalence. It reviews current early detection methods, such as Q-CHAT, and their limitations, and the role of machine learning in improving ASD diagnosis. Interpretability techniques like LIME are also discussed. The review highlights gaps in existing research and supports the study's methodology.

1.6.3 Chapter 3: Methodology

This chapter details the methods used to enhance ASD detection. It describes data preprocessing from the Q-CHAT dataset, the application of various machine learning, and model evaluation through cross-validation. The chapter also covers feature engineering and the use of interpretability techniques like LIME to ensure transparency and robustness.

1.6.4 Chapter 4: Results, Analysis and Discussion

This chapter presents the performance of machine learning models on the Q-CHAT dataset, including key metrics like accuracy and precision. It interprets these results, discusses model strengths and limitations, and explores the impact of interpretability techniques. The discussion connects findings to existing research and suggests directions for future work.

1.6.5 Chapter 5: Conclusion and Future Work

This chapter summarizes the key findings of the study on enhancing ASD detection in toddlers using machine learning. It highlights the improved diagnostic accuracy and efficiency achieved through various models and interpretability techniques. The chapter also discusses the implications of the results and outlines potential areas for future research, such as refining models and exploring new datasets to further advance early ASD detection.

2. Chapter Two: Literature Review

2.1 Background and Importance of ASD Detection in Toddlers Research

Autism Spectrum Disorder (ASD) is a significant neurodevelopmental disorder characterized by difficulties in social interaction, and communication, and the presence of restricted, repetitive behaviors or interests (Hirota and King, 2023). The early identification of ASD is critical since it has a considerable influence on a child's developmental trajectory and quality of life (Baghdadli et al., 2012). Recognizing ASD symptoms in toddlers is essential for implementing timely interventions that can lead to significant changes in developmental outcomes, such as improved social, cognitive, and linguistic abilities.

Early identification of ASD is associated with more successful therapeutic interventions and support, which can dramatically improve a child's capacity to function in daily life and prevent long-term impairment (Elder et al., 2017). The rising global prevalence of ASD highlights the critical need for better diagnostic approaches. Early detection goes beyond allowing effective early interventions to give families the tools and assistance they need to negotiate the obstacles associated with ASD (Grygas Coogle, Guerette, and Hanline, 2013). The focus on enhancing diagnostic accuracy and efficiency through advanced methodologies, including machine learning, is essential to improving the detection and management of ASD in toddlers.

2.2 Traditional Diagnostic Methods for Autism Spectrum Disorder (ASD) in Toddlers

Traditional diagnostic methods for Autism Spectrum Disorder (ASD) in toddlers involve a combination of developmental screenings, clinical evaluations, and standardized diagnostic tools (Hyman et al., 2020). Initially, developmental surveillance and routine screenings by pediatricians, such as the Modified Checklist for Autism in Toddlers (M-CHAT), help identify early signs of ASD (Eapen et al., 2013). If concerns are noted, a comprehensive diagnostic evaluation is conducted by a multidisciplinary team, including pediatricians, child psychologists, and speech therapists (Gerdtts et al., 2018). This evaluation involves direct observation of the child's behavior, detailed interviews with caregivers, and the use of standardized assessment tools like the Autism Diagnostic Observation Schedule (ADOS) and the Autism Diagnostic Interview-Revised (ADI-R) (Le Couteur et al., 2008). Additionally, behavioral and developmental checklists, such as the Childhood Autism Rating Scale (CARS) and the Social Communication Questionnaire (SCQ), are used to assess specific ASD-related behaviors (Magyar and Pandolfi, 2007). Neurodevelopmental assessments are also performed to differentiate ASD from other developmental disorders. Collaboration with early intervention services ensures that children receive personalized support based on diagnostic findings (Rutherford et al., 2021).

2.2.1 Limitations and Challenges of Traditional Diagnostic Methods for Autism Spectrum Disorder (ASD) in Toddlers

Traditional diagnostic methods for Autism Spectrum Disorder (ASD) in toddlers face several limitations and challenges. They can be time-consuming and resource-intensive, often resulting in delays in diagnosis and intervention (Lemay et al., 2020). The process is also subject to variability and subjectivity, as clinicians' interpretations and caregiver reports can introduce inconsistencies (Scheithauer et al., 2019). Standardized tools may not always provide perfect sensitivity and specificity, potentially leading to false positives or missed cases. Additionally, reliance on behavioral observations and caregiver input may not capture all early signs of ASD, and the one-size-fits-all approach may also be inappropriate to address individual differences effectively (Lesack et al., 2014). Accessibility issues further complicate the situation, with specialized assessments not always being readily available. These limitations highlight the need for more

efficient and precise diagnostic approaches, such as those involving machine learning technologies.

2.3 Scope of the Review

This review explores the application of various machine learning (ML) algorithms in the context of Autism Spectrum Disorder (ASD) detection in toddlers. The focus is on algorithms such as Logistic Regression, Support Vector Machine (SVM), K-nearest neighbors (KNN), Random Forest, Gradient Boosting Machines (GBM), Stacking Classifier, K-Mean Clustering, and Decision Tree along with Principal Component Analysis (PCA) for dimensionality reduction and ensemble methods. Each algorithm's theoretical background, benefits, limitations, and contributions to enhancing diagnostic accuracy and efficiency will be investigated. The review will examine existing studies and results that show the efficiency of various machine learning techniques in improving early identification and diagnosis of ASD and indicate areas that deserve additional exploration.

Additionally, the review will address the interpretability of ML models by examining methodologies like Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP). The aim is to emphasize the importance of making complex ML predictions transparent and understandable for healthcare professionals, thereby facilitating their integration into clinical practice. Overall, the review seeks to assess the effectiveness of current ML strategies in ASD detection, highlight existing challenges and limitations, and propose future research directions to advance the field.

2.4 AI and Machine Learning in ASD in Toddlers Research

2.4.1 Significance of Machine Learning in Detection of ASD in Toddlers

Machine learning (ML) holds significant promise for the early detection and diagnosis of Autism Spectrum Disorder (ASD) in toddlers, offering numerous advantages over traditional methods. By leveraging ML and deep learning (DL) architectures, clinicians can more accurately analyze and classify datasets to identify ASD traits, thus facilitating earlier and more precise interventions.

2.4.2 Enhanced Accuracy and Efficiency

Machine learning (ML) models, such as Support Vector Machine (SVM), k-nearest neighbors (KNN), and deep learning models like long short-term memory (LSTM), have demonstrated remarkable accuracy in classifying ASD traits. These advanced algorithms can process substantial amounts of data swiftly and efficiently, identifying subtle patterns that might be missed by human evaluators (Mukherjee, P., Sadhukhan, S., Godse, M. and Solutions, V.I., 2023). For instance, studies have shown that these models can achieve up to 100% accuracy in ASD detection by analyzing behavioral phenotypes and comorbidity patterns, significantly improving the reliability and speed of early diagnoses compared to traditional methods (Bone et al., 2016). This prominent level of accuracy ensures that children who exhibit early signs of ASD are identified and receive the necessary support timely.

2.4.3 Early Detection and Intervention

The ability of ML models to process and analyze large datasets promptly enables the possibility of earlier detection of ASD than conventional methods. Early identification is crucial because it allows for timely interventions that can improve a child's linguistic, cognitive, and social skills, leading to better developmental outcomes and reduced long-term healthcare burdens (Elder et al., 2017). With early and accurate detection, children can begin tailored therapeutic programs sooner, which can significantly enhance their overall development and quality of life (Elder et al., 2017). Early interventions have been demonstrated to improve communication skills, social interactions, and adaptive behaviors in children with ASD.

2.4.4 Improvement in Clinical Workflow

Machine learning (ML) significantly improves clinical workflows in detecting Autism Spectrum Disorder (ASD) in toddlers by automating routine tasks, enhancing diagnostic accuracy, and streamlining care delivery (Aylward et al., 2023). ML automates administrative tasks such as transcribing clinical encounters, reducing the workload on healthcare providers (Coiera et al., 2018). It processes large datasets to identify subtle patterns indicative of ASD, enabling earlier and more accurate diagnosis compared to traditional methods. ML also shortens the time required for diagnostic screenings and differentiates between ASD and other developmental disorders like ADHD with high accuracy (Abbas et al., 2020; Tariq et al., 2018). Additionally, ML integrates comprehensive patient data to develop personalized treatment plans and facilitates better

communication among healthcare providers (Buch et al., 2018). These advancements lead to more efficient healthcare delivery, reduced clinician workload, and improved patient outcomes.

2.5 Types of Machine Learning

Machine learning is divided into three categories: supervised, unsupervised, and reinforcement learning.

2.5.1 Supervised Learning

Supervised learning involves training ML algorithms on labeled datasets, where each input is paired with the correct output. In the detection of ASD, supervised learning can be particularly effective. For instance, algorithms are trained on datasets that include labeled examples of toddlers with and without ASD, along with their clinical features and behavioral data (Hyde et al., 2019). By learning from these labeled examples, the model can identify patterns and make accurate predictions about new, unseen data. This method is useful for tasks such as classifying whether a toddler exhibits behaviors consistent with ASD or predicting the likelihood of ASD based on various diagnostic criteria (Mohanty et al., 2021). Supervised learning helps in developing predictive models that can flag potential ASD cases early, leading to timely intervention and improved outcomes (Baadel et al., 2020).

2.5.2 Unsupervised Learning

Unsupervised learning deals with unlabeled data and aims to find hidden patterns or structures within the data (Igual, L., Seguí, S., 2017). In the context of ASD detection, unsupervised learning can be employed to identify natural groupings or clusters in behavioral data or electronic medical records (EMRs) without predefined categories (Alzoubi et al., 2019). For example, unsupervised learning algorithms might reveal distinct clusters of symptoms or behavioral patterns associated with ASD, which can then inform the development of new diagnostic criteria or screening tools (Stevens et al., 2019). This approach is also useful for discovering novel subtypes of ASD or comorbid conditions that were not previously recognized, contributing to a more nuanced understanding of the disorder (Beverdors, 2016).

2.5.3 Reinforcement Learning (RL)

Reinforcement learning focuses on training algorithms through interactions with an environment to maximize cumulative rewards (Oh et al., 2020). In terms of ASD detection, reinforcement

learning could be applied to optimize diagnostic processes or treatment strategies. For instance, an algorithm might learn to make diagnostic decisions by receiving feedback on its accuracy, thereby improving its performance over time (Shin et al., 2019). This approach could also be used to develop personalized intervention plans by continuously adjusting recommendations based on the effectiveness of previous treatments and patient responses. While less commonly used for initial diagnosis, reinforcement learning holds the potential for refining and personalizing ongoing management and therapeutic approaches for toddlers with ASD (Dekhil et al., 2018).

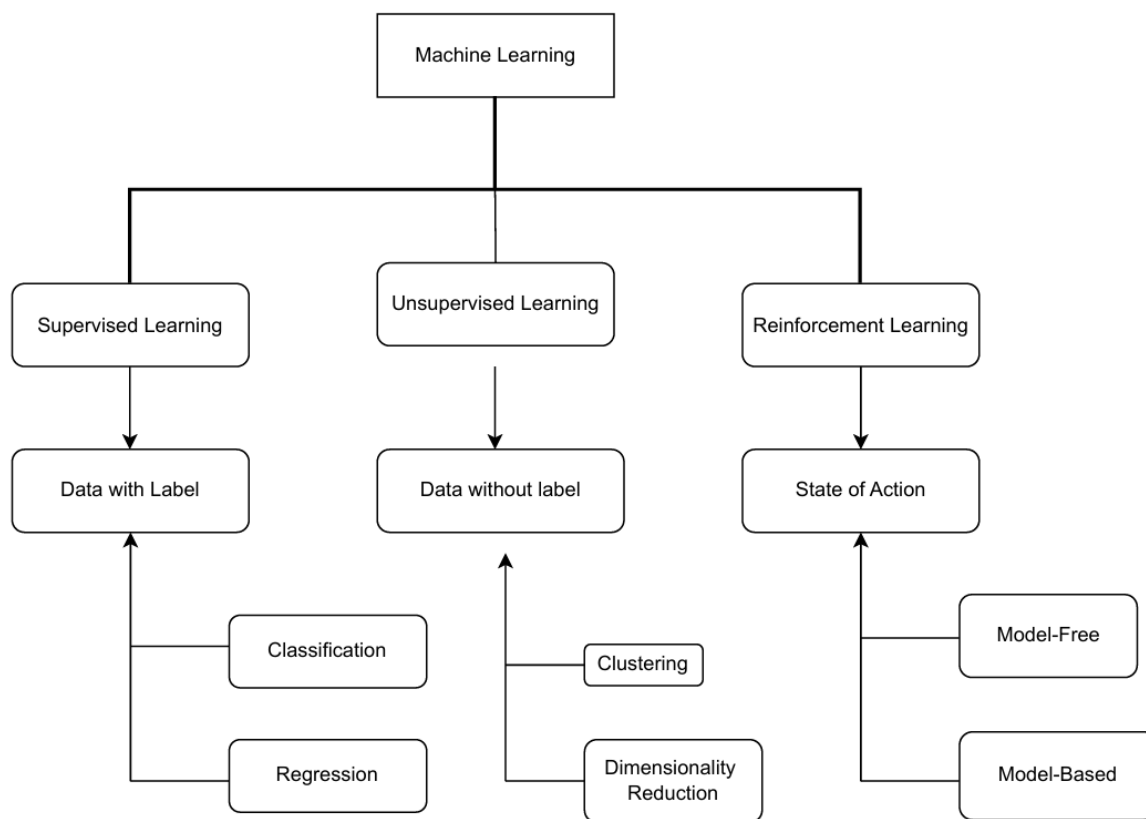


Figure 1: Categories of Machine Learning

2.6 Machine Learning Techniques for Detection of Autism in Toddlers: Enhanced Algorithms for Diagnosis and Risk Prediction

The integration of machine learning (ML) techniques into autism research has significantly advanced the early detection, accurate diagnosis, and effective risk assessment of Autism Spectrum Disorder (ASD) in toddlers (Vakadkar et al., 2021). This section explores various ML

techniques, which include Support Vector Machine (SVM), K-nearest neighbors (KNN), Random Forest, Gradient Boosting Machines, Logistic Regression, Decision Tree, and Stacking Classifier, and examines their specific applications in detecting ASD in toddlers.

2.6.1 Logistic Regression (LR)

To predict the probability of a target variable, statisticians use Logistic Regression (LR), a supervised learning classification method. In this approach, the dependent variable is binary, meaning it can only be classified as either 1 (for success) or 0 (for failure). With an LR model, it is possible to predict the probability $P(Y = 1)$ for a given value of X . The most basic form of LR is known as binary or binomial logistic regression, which assumes that the dependent or target variable can only take on the values 1 or 0 (Talukdar, Gogoi, & Singh, 2023). This allows us to model the relationship between a binary outcome and a set of predictors. In LR, the linear function is primarily used as an input to another function, such as g , as shown in the following relationship:

$$h_{\theta}(x) = g(\theta^T x) \text{ where } 0 \leq h_{\theta} \leq 1 \quad 1$$

Studies on ASD detection in toddlers highlighting Logistic Regression

In a study by Kosmicki et al., Logistic Regression (LR) was used to analyze ADOS Module 2 records to detect autism in toddlers. The analysis was conducted on 362 autistic and 282 non-autistic individuals, using backward feature selection and managing class imbalance with weighted classes. The optimal LR model, featuring L1 regularization and nine features, was evaluated on an independent dataset of 1,089 autistic and 66 non-autistic individuals. The model achieved a high sensitivity of 98.8% and specificity of 89.4%, with an overall unweighted average recall (UAR) of 94.1%. This study underscores the potential of logistic regression to enhance ASD detection in toddlers, thereby improving clinical decision-making and outcomes for early intervention (Kosmicki et al., 2015). In another study by Talukdar, Gogoi, and Singh, Logistic Regression (LR) was employed for ASD detection in toddlers. The performance metrics for LR were as follows: the classification accuracy was 68.13%, precision was 68.13%, recall was 100%, the F1-score was 81.05%, and the area under the curve (AUC) was 98.4%. These results demonstrate the effectiveness of Logistic Regression in identifying ASD in toddlers, with particularly high recall and AUC values, indicating strong sensitivity and discriminatory power (Talukdar, Gogoi, and Singh, 2023).

2.6.2 Support Vector Machine (SVM)

Support Vector Machines (SVM) are supervised machine-learning techniques used for both classification and regression tasks. The primary goal of the SVM technique is to identify a hyperplane in an N-dimensional space that can classify the data points. To distinguish between classes, SVM generates a decision threshold. The Radial Basis Function (RBF) kernel is versatile in managing both linearly separable and complex non-linear datasets (Talukdar, Gogoi, & Singh, 2023). The RBF kernel function calculates the similarity or distance between two points, X_1 and X_2 . The mathematical representation of the kernel is as follows:

$$K(X_1, X_2) = \exp\left(-\frac{\|X_1 - X_2\|^2}{2\alpha^2}\right) \quad 2$$

where α is the variance and the hyperparameter $\|X_1 - X_2\|$ represents the Euclidean (L2-norm) distance between two points, X_1 and X_2 .

Studies on ASD detection in toddlers highlighting SVM.

In a study by Talukdar, Gogoi, and Singh (2023) Support Vector Machines (SVM) model was explored, and the classification accuracy achieved was 77.28%, with a precision of 75%. The model demonstrated a recall of 100%, an F1-score of 85.71%, and an area under the curve (AUC) of 99.4%. These metrics highlight SVM's effectiveness in detecting ASD in toddlers, showing high accuracy, perfect recall, and excellent discriminatory power (Talukdar, Gogoi, and Singh, 2023). In Squarcina et al. (2021), MRI data of high quality was analyzed using Radial Basis Function (RBF) SVM with 68 Regions of Interest (ROIs) from the Desikan-Killiany FreeSurfer atlas. Initial results showed that five ROIs achieved accuracies above 70%, with the right hemisphere superior frontal thickness reaching 77.6% accuracy, 83.3% specificity, and 72.5% sensitivity. Following this, a recursive feature selection process identified the optimal combination of eight ROIs, resulting in the highest accuracy of 84.2%. These ROIs included regions from both hemispheres, such as the right hemisphere superior frontal gyrus and left hemisphere middle temporal gyrus (Squarcina et al., 2021). These results underscore SVM's strong potential for accurate ASD detection and improved diagnostic capabilities.

2.6.3 Random Forest (RF)

Random Forest (RF) is a classification method that constructs an ensemble of decision trees to classify data. The RF model generates a proximity matrix that reflects the similarity between participants. This matrix shows how frequently pairs of subjects are grouped into the same terminal node across the decision trees and resembles a correlation matrix. Conceptually, this proximity matrix can be viewed as a graph, where a community detection algorithm (Rosvall and Bergstrom, 2008; Gates et al., 2016) can be applied to identify potential subgroups. Several recent studies have employed community detection to characterize subpopulations (Fair et al., 2012a). However, a limitation of the current application of this approach is that community detection is not directly linked to the outcome measurement of interest.

Mathematically: Given B trees, predictions for new data x' are computed as:

$$\hat{f}(x') = \frac{1}{B} \sum_{b=1}^B f_b(x') \quad 3$$

Studies on ASD detection in toddlers highlighting Random Forest

In the study by Feczko et al., Random Forest (RF) was used to classify individuals with Autism Spectrum Disorder (ASD) from controls based on behavioral data from seven tasks. The RF model achieved a classification accuracy of 73%, significantly outperforming the permutation accuracy of 51%. It also demonstrated a sensitivity of 63% and a specificity of 81%, both of which were notably higher than their respective permutation measures. Additionally, a secondary RF analysis, accounting for age and gender, confirmed that the model still effectively distinguished ASD from control participants, despite these confounding factors (Feczko et al., 2018). In the study by Talukdar, Gogoi, and Singh, Random Forest (RF) achieved a classification accuracy of 93.69% for detecting Autism Spectrum Disorder (ASD). The model demonstrated high precision at 91.52%, perfect recall at 100%, an F1-score of 95.57%, and an impressive area under the curve (AUC) of 99.7% (Talukdar, Gogoi, and Singh, 2023). These results highlight the RF model's exceptional performance in accurately identifying ASD.

2.6.4 Decision Tree (DT)

Decision Trees are supervised learning algorithms primarily employed for classification tasks. They structure information hierarchically, with a root node, intermediate branches, and terminal

leaves representing features, decisions, and outcomes, respectively (Sheth, Tripathi, and Sharma, 2022). By partitioning data at each node to distinguish between classes, these algorithms provide a straightforward visual framework for evaluating different decision outcomes based on existing conditions.

Mathematically, the decision tree algorithm uses an impurity measure like Gini Index or Entropy to split the data. For a node t in the tree, the Gini Index is calculated as:

$$G(t) = 1 - \sum_{i=1}^C p_i^2 \quad 4$$

where p_i is the proportion of samples belonging to class i at node t , and C is the total number of classes. The split that results in the lowest Gini Index is chosen to partition the data.

Studies on ASD detection in Toddlers Highlighting Decision Tree

The study by Cohen and Flory (2019) reveals that the Decision Trees (DT) model can be efficient in classifying ASD subgroups in toddlers. By leveraging PDDBI T-scores, the DT model successfully identified distinct subgroups—MINVERB, ATYPICAL, and VERBAL—based on sensory behaviors, repetitive language, and other domains. The classification effectively highlighted differences in seizure history, VABS scores, and cognitive ability among the subgroups, even though sex ratios were consistent across groups. These findings underscore the utility of Decision Trees in distinguishing between ASD subgroups, offering valuable insights into the variability of ASD presentations and aiding in tailored diagnostic and intervention strategies (Cohen and Flory, 2019).

2.6.5 K-Means Clustering

K-means clustering is an unsupervised machine learning algorithm used to detect autism spectrum disorder (ASD) in toddlers by grouping behavioral data into K clusters. The algorithm selects initial centroids, assigns each toddler's data to the nearest centroid, and recalculates the centroids based on the mean of the assigned data points. This process repeats until the centroids stabilize. The resulting clusters help identify groups of toddlers with similar behaviors, aiding in the detection of those with a higher likelihood of ASD for further evaluation and early intervention (Parlett-Pelleriti et al., 2023).

Studies on ASD detection in toddlers highlighting K-Means Clustering

In 2016, Lingren and colleagues used data from 20,658 subjects with ASD and other comorbid disorders to create 4-5 clusters using k-means clustering and standardized PheWAS codes. Each of the three sites studied produced similar clusters. Cluster 1 included psychiatric issues like anxiety and depression, Cluster 2 comprised developmental disorders such as dyslexia and coordination problems, and Cluster 3 was associated with epilepsy. The largest cluster did not show a high occurrence of specific comorbidities. (Lingren et al., 2016). Hu and Steinberg (2009) used 123 ADI-R item scores from 1,954 subjects to cluster ASD phenotypes and identify more homogeneous subgroups for gene expression analysis. They combined principal component analysis, hierarchical clustering, and k-means clustering, concluding that four clusters best represent ASD: severe language deficits, milder symptoms, notable savant skills, and intermediate severity. Male subjects from the first three clusters were further analyzed for gene expression (Hu and Steinberg, 2009).

2.6.6 Ensemble Methods

Ensemble methods in ASD detection involve combining multiple machine-learning models to enhance prediction accuracy and robustness (Bala et al., 2022). These methods aggregate the outputs of different models, which can be of different types, to create a more reliable and accurate system for identifying ASD. Common techniques include bagging, which builds and aggregates predictions from multiple models trained on different data subsets, boosting, which sequentially builds models to correct errors from previous ones, and stacking, which uses a meta-learner to combine predictions from multiple models. In ASD detection, ensemble methods improve classification accuracy, integrate diverse data sources such as behavioral and genetic information, and handle the complexity of ASD symptoms by leveraging the strengths of various models.

2.7 Comparative Study Assessing Machine Learning Methods' Effectiveness in Toddler ASD Detection

Comparative studies on machine learning (ML) algorithms for autism spectrum disorder (ASD) detection evaluate their effectiveness in terms of accuracy, sensitivity, and specificity.

The study by Safahi et al. (2024) assessed four advanced machine learning classifiers—Neural Networks, Random Forests, Support Vector Machines, and Gradient Boosted Models—for

detecting autism spectrum disorder (ASD) in toddlers and adults. Using features selected through step forward selection (SFS), the models achieved perfect accuracy, precision, recall, F1-scores, and AUC-ROC for toddlers, and over 99% accuracy with near-perfect precision and recall for adults (Safahi et al., 2024).

Similarly, Shrivastava and colleagues explores machine learning (ML) for early autism spectrum disorder (ASD) detection using the INDT-ASD Indian database. Various ML classifiers were tested, such as Adaboost, Gradient Boost, Decision Tree, Logistic Regression, Random Forest, Gaussian Naive Bayes, Linear Discriminant Analysis, Quadratic Discriminant Analysis, K-Nearest Neighbors, with Support Vector Machine (SVM) achieving the highest accuracy of 100%. The study reduced the number of diagnostic questions from 28 to 20 and introduced a bilingual web-based solution, improving accessibility and efficiency in ASD detection (Shrivastava, T., Chaudhari, H. and Singh, V., 2024).

Florio et al. (2009) evaluated the accuracy of a neural network model for detecting autism spectrum disorder (ASD) using the Developmental Behavior Checklist (DBC). The model, tested on 319 ASD and 319 non-ASD developmental disorder (DD) cases, achieved 92% accuracy and an AUC of 0.93. In cross-validation with a separate sample of 62 ASD and 38 non-ASD DD cases, it yielded 80% accuracy, 92% sensitivity, 70% specificity, and an AUC of 0.88. This study contributes to understanding machine learning's effectiveness in ASD detection (Florio et al., 2009).

Abbas et al. (2018) developed an affordable and efficient ASD screening tool using Random Forest (RF) classifiers. They combined two RF classifiers: one based on a parental questionnaire and another on behaviors observed in home videos. The parental classifier was trained on data from 2299 children with ASD, 100 typically developing (TD) children, and 287 with other conditions, while the video classifier was trained on 3310 ASD children, 585 TD children, and 364 with other conditions. The classifiers were trained separately for children younger and older than 4 years due to differences in language development. The final diagnosis was determined using logistic regression on the results from both classifiers. This method outperformed established tools like the Modified Checklist for Autism in Toddlers (MCHAT) and the Child Behavior Checklist (CBCL) based on ROC curves (Abbas et al., 2018).

Additionally, Duda et al. (2016) developed a reduced-item classifier for autism spectrum disorder (ASD) using the Social Responsiveness Scale (SRS) and machine learning models. They trained

and tested six models on data from 2775 individuals with ASD and 150 with ADHD. Using forward feature selection, under sampling, and 10-fold cross-validation, four models—SVM, ridge regression, LASSO, and linear discriminant analysis—performed well, each relying on just five of the 65 SRS items. These models achieved an AUC between 0.962 and 0.965, demonstrating high accuracy in distinguishing ASD from ADHD (Duda et al., 2016).

Jiao et al. (2012) employed machine learning to classify children with autism spectrum disorder (ASD) based on symptom severity using single nucleotide polymorphism (SNP) data. The study analyzed SNP data from 118 children aged 1.5 to 14 years and categorized them into mild/moderate or severe groups using the Childhood Autism Rating Scale (CARS). Decision stumps and FlexTrees were identified as the most effective models, achieving 67% accuracy, 88% sensitivity, and 42% specificity. FlexTrees, which enhance binary-tree methods by combining multiple features, and specific SNPs demonstrated notable predictive power for ASD symptom severity. The results indicate that SNPs can moderately predict ASD symptom severity (Jiao et al., 2012).

These studies collectively highlight the significant potential of machine learning (ML) in advancing autism spectrum disorder (ASD) detection. They provide insights into the effectiveness of various algorithms and their capabilities in identifying ASD across different age groups and data types. As ML technology continues to evolve, its application in ASD research promises to enhance diagnostic accuracy and early detection, paving the way for more personalized and effective approaches to autism care.

Table 1: Summary of Related Literature on ML Techniques for ASD Detection in Toddlers

Author	Algorithm(s) Used	Dataset Used	Contribution	Limitation
Safahi et al. (2024)	Neural Networks, Random Forests, Support Vector Machines, Gradient Boosted Models	Features from Step Forward Selection (SFS) dataset	The study achieved 100% accuracy, precision, recall, and AUC-ROC in toddler ASD detection	The study is limited to a specific dataset; generalizability needs further validation

Shrivastava et al. (2024)	Adaboost, Gradient Boost, Decision Tree, Logistic Regression, Random Forest, Gaussian Naive Bayes, LDA, QDA, KNN, SVM	This paper utilized the dataset from INDT-ASD Indian Database	This research had improved accuracy with reduced diagnostic questions and introduced a bilingual web-based solution	The results are likely to vary with different datasets and populations
Florio et al. (2009)	Neural Network	Developmental Behavior Checklist (DBC) dataset	This research achieved 92% accuracy and AUC of 0.93; cross-validation showed 80% accuracy	Lower specificity: cross-validation sample size was small
Abbas et al. (2018)	Random Forests (RF)	Parental questionnaires and home videos dataset	The paper developed a dual RF classifier system outperforming traditional tools like MCHAT and CBCL	The developed model may not be able to account for all developmental variances and conditions
Duda et al. (2016)	SVM, Ridge Regression, LASSO, Linear Discriminant Analysis	Social Responsiveness Scale (SRS)	The analysis achieved high AUC (0.962 to 0.965) with reduced-item classifiers	The result is limited to data from individuals with ASD and ADHD
Jiao et al. (2012)	Decision Stumps, FlexTrees	SNP data from Childhood Autism Rating Scale (CARS)	The paper demonstrated the predictive power of SNPs for classifying ASD severity	Moderate specificity: the effectiveness may vary with different symptom of severities

2.8 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a technique for reducing the dimensionality of a dataset while retaining as much information as possible. It transforms the original variables into new, uncorrelated variables called principal components. These components capture the maximum variance and are orthogonal to each other. PCA simplifies the dataset by replacing groups of correlated variables with fewer principal components, making high-dimensional data easier to visualize and analyze (Shihab, Dawood and Kashmar, 2020).

2.9 Improving Interpretability of ML Models in Autism Detection: Focus on Toddler ASD

Pandya and Jain (2024) focus on enhancing the interpretability of machine learning (ML) models used for autism spectrum disorder (ASD) detection. They emphasize the use of Explainable Artificial Intelligence (XAI) methods, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), to improve transparency in how ML models make predictions. Their study highlights the importance of these techniques in making complex models more understandable for clinicians, aiming to foster trust and facilitate the adoption of ML in autism diagnosis. They also advocate for ongoing advancements in interpretability tools, including the integration of domain-specific knowledge and real-time model monitoring (Pandya & Jain, 2024).

2.9.1 Need for Model Interpretability

Model interpretability is essential in autism spectrum disorder (ASD) detection to translate complex machine learning (ML) outputs into actionable clinical insights. As ML tools become more integrated into diagnostic processes, understanding how these models make decisions is crucial for building clinician trust, meeting regulatory and ethical standards, and ensuring fairness in diagnoses. Interpretability helps identify and address biases, facilitates model debugging and improvement, and enhances communication with patients and families by making results more comprehensible. Techniques like SHAP and LIME are used to provide transparency into feature importance and decision-making processes, helping to align advanced ML models with practical clinical needs and ensuring their effective and responsible application in ASD detection.

2.9.2 Techniques for Enhancing Interpretability- Local Interpretable Model-agnostic Explanations (LIME)

Local Interpretable Model-agnostic Explanations (LIME) enhances the interpretability of machine learning models by focusing on individual predictions. It generates perturbed data around a specific instance and fits a simple model to these points to approximate the complex model's behavior locally. This approach is crucial for autism detection in toddlers as it provides insights into which features influence predictions, helping clinicians validate and understand the model's decisions. Recent advancements, including decision rules and techniques like Leave-One-Covariate-Out (LOCO) and Local Rule-based Explanation (LORE), further improve interpretability by offering detailed explanations and assessing feature importance (Zafar, Khan, and Rehman, 2021).

2.9.3 Techniques for Enhancing Interpretability- SHAP (SHapley Additive exPlanations)

SHAP (SHapley Additive exPlanations) improves model interpretability by using Shapley values from game theory to measure each feature's contribution to a model's prediction. Developed by Lloyd Shapley in 1953, this method evaluates how much each feature impacts the prediction, considering all possible combinations of features. The SHAP value for a feature is calculated by examining its effect in various subsets of features, ensuring a fair and comprehensive assessment. This approach helps clarify which features are most influential in autism detection models, enhancing transparency and trust in the model's predictions (Sahlaoui et al., 2021).

2.10 Case Studies of LIME and SHAP Application in ASD Detection

Sujana & Augustine (2023): In their study, Sujana and Augustine developed an interpretable autism diagnosis model using the ABIDE dataset, which included 1,114 records. The neural network model achieved an accuracy of 91.37%. To improve the interpretability of this model, the authors employed Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), and ANCHOR. These tools provided clear and visual explanations for the model's predictions, enhancing the transparency of its decision-making process. The effective application of LIME and SHAP in this context showcased their ability to make complex models more understandable and reliable, thus supporting their use in clinical settings for autism diagnosis (Sujana & Augustine, 2023).

2.11 Variable Importance in Autism Detection Models

The advancement of machine learning (ML) techniques in autism detection has emphasized the importance of identifying and selecting key predictive features. Features like behavioral scores and developmental milestones play a key role in predictive models' effectiveness. By focusing on these important variables, researchers can enhance the accuracy and reliability of ML models for autism diagnosis, leading to more personalized and effective assessments.

2.12 Feature Selection

Feature selection is a critical step in the machine learning process that involves identifying and selecting the most relevant features (or variables) from a dataset to improve the performance of predictive models. The goal is to reduce the complexity of the model, enhance its accuracy, and avoid overfitting by eliminating irrelevant or redundant features. Proper feature selection can significantly impact the performance of a machine learning model by focusing on the most informative features and ignoring those that do not contribute meaningfully to predictions.

2.12.1 Filter Methods

Filter methods evaluate features based on their inherent data characteristics, such as Pearson's correlation or chi-square tests, without involving classification algorithms. These methods select features independently of the classifier and do not consider how the selected features might affect the performance of the classification model (Al-Tashi et al., 2020).

2.12.2 Wrapper Methods

Wrapper methods, in contrast, use a specific classifier to assess feature subsets. They iteratively refine the feature set to improve model performance, with techniques like recursive feature elimination being common examples. Although more computationally intensive, wrapper methods tailor feature selection to the classifier's performance (Al-Tashi et al., 2020).

2.12.3 Embedded Methods

Embedded methods integrate feature selection directly into the model training process. Algorithms like LASSO (Least Absolute Shrinkage and Selection Operator) and Random Forest include built-in feature selection capabilities (Rahman et al., 2020). These methods combine the strengths of

both filter and wrapper methods by making feature selection part of the model training, resulting in a more efficient and effective process.

2.13 Findings from the Literature Review on Machine Learning in Autism Detection and Diagnosis

The literature review reveals the significant potential of machine learning (ML) techniques in detecting and diagnosing autism spectrum disorder (ASD) in toddlers, with algorithms like Random Forests and Support Vector Machines showing promise in early screening, especially for toddlers. Despite achieving high accuracy, challenges remain in ensuring the generalizability and interpretability of these models, which are crucial for their adoption in clinical settings. Recent advancements have focused on enhancing model transparency through tools like SHAP and LIME, increasing clinician trust. Feature selection is also highlighted as essential for improving model performance and usability, particularly when using diverse data sources like the Q-CHAT dataset. Building on these findings, this research aims to further enhance ML models' interpretability and clinical applicability for toddler ASD detection by refining feature selection processes and employing advanced interpretability techniques. The goal is to develop diagnostic tools that are accurate and provide actionable insights for clinicians, addressing key challenges identified in the existing literature.

2.14 Refined Research Questions

Based on the literature review and the challenges identified in autism detection using machine learning, the refined research questions for this study are:

a) How do various machine learning models perform in detecting Autism Spectrum Disorder (ASD) in toddlers using the Q-CHAT dataset?

This question aims to compare the performance of different machine-learning models for ASD detection.

b) What is the impact of feature selection using Recursive Feature Elimination (RFE) on the performance of machine learning models for ASD detection?

This question focuses on assessing how feature selection improves model performance for ASD detection.

- c) **What are the most important features identified by machine learning models in predicting ASD risk using the Q-CHAT dataset?**

This question seeks to identify key features for predicting ASD risk.

- d) **How do Local Interpretable Model-agnostic Explanations (LIME) contribute to the interpretability of machine learning models for ASD detection?**

This question explores how LIME enhances the interpretability of ASD detection models.

By addressing these research questions, this study will enhance autism detection in toddlers, improving model accuracy, and interpretability, thus supporting better early diagnosis and intervention.

3. Chapter Three: Methodology

This chapter describes the methodology for detecting autism spectrum disorder (ASD) in toddlers using the Q-Chat dataset and machine learning (ML) techniques. The aim is to improve diagnostic accuracy and model interpretability while addressing data quality and bias challenges.

3.1 Research Approach

This study employs a quantitative approach using the Q-Chat dataset to build and evaluate ML models for ASD detection. The methodology includes data preprocessing, feature selection, model development, and evaluation.

3.1.1 Experimental Framework

The experimental framework for detecting autism spectrum disorder (ASD) in toddlers using machine learning (ML) is designed to systematically evaluate and improve the effectiveness of various ML techniques. This framework includes several key components: data management, model development, evaluation, and interpretability. Each component is tailored to address the specific challenges identified in previous research and aims to enhance the accuracy and practical application of ML models in clinical settings.

This methodology is depicted in Figure 2, which illustrates the sequential steps involved in the process:

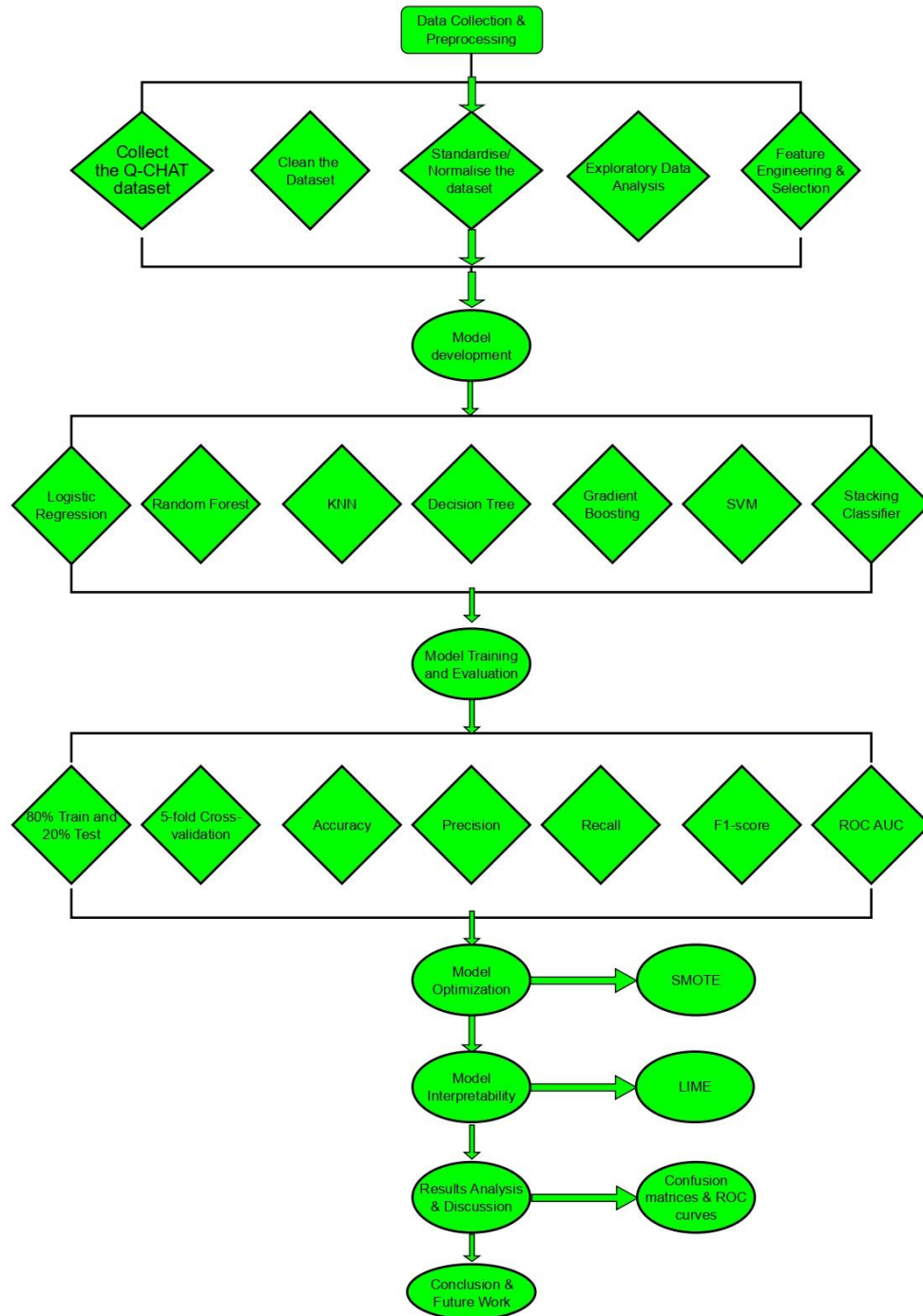


Figure 2: Comprehensive Workflow for ML Model Development in Detection of ASD Research

This flowchart provides a comprehensive overview of the methodological steps, ensuring a systematic approach to improving machine learning models for autism detection in toddlers.

3.2 Implementation Tools and Software

The analysis for detecting autism spectrum disorder (ASD) in toddlers utilizes Python for its rich array of data science and machine learning libraries. Key tools include Pandas for data manipulation, Scikit-learn for implementing and evaluating machine learning models using metrics such as accuracy, precision, recall, F1 score, AUC-ROC, and confusion matrix, and SMOTE for addressing class imbalance. Visualization and performance metrics are handled with Matplotlib, NumPy, and Seaborn. Feature selection and dimensionality reduction are performed using PCA and K-Means clustering. For interpretability, LIME is employed to clarify model predictions. The entire process is conducted within Jupyter Notebooks, providing a flexible and interactive environment for coding and analysis.

3.3 Expected Outcomes

This research on detecting autism spectrum disorder (ASD) in toddlers using machine learning is expected to significantly enhance diagnostic accuracy through refined models and advanced feature selection techniques. It will improve the interpretability of these models with tools like LIME, making the decision-making processes more transparent and actionable for clinicians. The research aims to identify key predictive features for ASD, address biases, and ensure that the models generalize well across diverse populations. Ultimately, it will contribute to the integration of machine learning into clinical practice, providing tools that can improve early detection and diagnosis of ASD.

3.4 Ethical Considerations

The use of machine learning (ML) in detecting autism spectrum disorder (ASD) in toddlers involves several ethical considerations that must be carefully addressed to ensure responsible and fair use of technology.

3.4.1 Data Privacy and Confidentiality: As machine learning models increasingly rely on sensitive health data, ensuring data privacy and security becomes vital. This dissertation uses anonymized datasets to protect patient identities, but further measures are necessary when deploying these models in real-world settings. Data encryption should be implemented to

safeguard patient information both at rest and in transit. Additionally, techniques like federated learning could be explored, where models are trained across decentralized data sources without the need for raw data to leave local systems. This approach minimizes the risk of data breaches and aligns with stringent data protection regulations like GDPR.

3.4.2 Informed Consent and Transparency: Transparency in AI is crucial for maintaining patient trust and ensuring ethical use of technology in healthcare. The use of interpretability tools like LIME in this dissertation is a step toward making the model's decisions understandable to clinicians and patients alike. However, transparency must also extend to the consent process. Patients should be informed when AI is being used as part of their diagnostic process, with clear explanations provided about how the model works and what role it plays in their diagnosis. This transparency helps patients make informed decisions about their care and fosters trust in the healthcare system.

3.4.3 Bias and Fairness in AI: The risk of bias in machine learning models, particularly in healthcare, cannot be ignored. This dissertation acknowledges the potential for bias in the dataset used, given its demographic composition. To mitigate this, future research should focus on creating more diverse and representative datasets, which include data from various demographic groups to ensure the model's fairness and generalizability. Furthermore, bias detection tools should be integrated into the model development process. By regularly auditing the model's performance across different subgroups, we can identify and address any disparities. This would help ensure that the model's predictions are equitable and do not disproportionately disadvantage any group.

3.4.4 Impact on Decision-Making: These ML models are intended to support, not replace, clinical judgment. Results must be interpreted with professional expertise to avoid over-reliance on technology.

3.5 Design and Implementation

The study begins with a detailed Exploratory Data Analysis (EDA) of the Q-Chat dataset (Niedźwiecka and Pisula, 2022) to understand its features and distribution of ASD diagnoses. This phase informs the subsequent design and implementation of predictive models. The study utilized various machine learning algorithms, including Random Forest, SVM, Gradient Boosting, Decision Trees, K-Nearest Neighbor, and Logistic Regression, to analyze the Q-CHAT dataset. These models were selected for their complementary strengths in handling complex data and

improving interpretability and accuracy in toddlers' ASD detection. Additionally, a stacking model was employed to combine the predictions of these algorithms, enhancing overall performance and reliability. This approach aimed to maximize predictive accuracy and generalizability, contributing to more effective and clinically applicable ASD diagnostic tools. The models are rigorously evaluated using metrics such as accuracy, precision, recall, ROC AUC, and F1 score, complemented by ROC curves and confusion matrices.

To enhance model performance and handle class imbalance, techniques like SMOTE (Synthetic Minority Over-sampling Technique), K-Means clustering, and Principal Component Analysis (PCA) are applied. Recursive Feature Elimination (RFE) is used for refined feature selection, while interpretability tools like LIME are utilized to improve model transparency and trustworthiness. This approach aims to develop accurate and interpretable models for ASD detection, contributing to better diagnostic and treatment outcomes for toddlers.

3.6 Exploratory Data Analysis (EDA) and Processing

3.6.1 Dataset Overview

The dataset used in this study consists of information from 1,016 participants, categorized into four groups: typically developing (TD) children, children with ASD concerns, children with siblings who have ASD (ASD-sibling group), and children with developmental delays (DD group). The dataset comprises of various features, such as ‘child_ID’, ‘sex’, ‘age’, ‘group’, ‘asd_risk’, ‘sum_qchat’, and a series of responses to Q-Chat questions (‘Q01’ to ‘Q25’). Each row represents individual records, with ‘child_ID’ serving as a unique identifier.

The dataset features demographic information such as ‘sex’, ‘age’, and ‘group’, along with ‘asd_risk’, which indicates the risk level of ASD. ‘sum_qchat’ is a numerical summary of responses, and Q01 through Q25 represent responses to specific questions within the Q-Chat questionnaire. This comprehensive set of variables allows for an in-depth analysis of ASD risk based on questionnaire responses and demographic factors.

3.6.2 Variables Description

Table 2: Variables Description of the Q-Chat Dataset

Variable	Description
----------	-------------

child_ID	A unique identifier for each child, used to distinguish individual records.
sex	The gender of the child represented numerically (e.g., 1 for male, 2 for female).
age	The child's age at the time of assessment, typically recorded in months.
group	The diagnostic group classification for the child (e.g., TD, ASD-concerns, ASD-sibling, DD)
asd_risk	Indicates the child's risk level for autism spectrum disorder (e.g., 2 for high risk, 1 for low risk).
sum_qchat	Total score from the Q-Chat assessment, summarizing responses to various questions related to ASD risk.
Q01	Look when call name: Measures if the child looks when their name is called.
Q02	Eye contact: Measures the child's frequency of making eye contact.
Q03	Line objects up: Measures the child's tendency to line up objects.
Q04	Understand child's speech: Measures how well the child understands spoken language.
Q05	Protoimperative pointing: Measures the child's use of pointing to request objects or actions.
Q06	Protodeclarative pointing: Measures the child's use of pointing to share interest or draw attention.
Q07	Interest maintained by spinning object: Measures the child's sustained interest in spinning objects.
Q08	Number of words: Measures the total number of words the child uses.
Q09	Pretend play: Measures the child's engagement in pretend or imaginative play.
Q10	Follow a look: Measures the child's ability to follow the direction of someone else's gaze.
Q11	Sniff/lick unusual objects: Measures if the child sniffs or licks objects that are unusual.
Q12	Use of hand as tool: Measures if the child uses their hand to manipulate objects in a functional way.
Q13	Walk on tiptoes: Measures if the child walks on their tiptoes frequently.

Q14	Adapt to change in routine: Measures the child's ability to adapt to changes in their routine.
Q15	Offer comfort: Measures if the child offers comfort to others.
Q16	Do same thing over and over again: Measures if the child engages in repetitive behaviors.
Q17	Typicality of first words: Measures the typicality of the child's first spoken words.
Q18	Echolalia: Measures if the child repeats words or phrases they have heard.
Q19	Gestures: Measures the child's use of gestures for communication.
Q20	Unusual finger movements: Measures if the child exhibits unusual finger movements.
Q21	Check reaction: Measures if the child checks the reaction of others during interactions.
Q22	Maintenance of interest: Measures if the child maintains interest in activities or objects.
Q23	Twiddle objects repetitively: Measures if the child twiddles or fiddles with objects repetitively.
Q24	Oversensitive to noise: Measures if the child is unusually sensitive to sounds or noises.
Q25	Stare at nothing with no purpose: Measures if the child stares at nothing in a purposeless manner.

3.6.3 Descriptive Statistics

Descriptive statistics, illustrated in Figure 3, provide essential insights into the distributions, variability, and quality of the features within the dataset. These statistics are crucial for understanding the data, facilitating comparisons, and effectively communicating key characteristics of the dataset.

```

Descriptive statistics for numeric variables:
  Variable  Count    Mean    Std   Min   25%   50%   75%   Max
0      sex  1016.0   1.409449  0.491974  1.0    1.0    1.0    2.0    2.0
1      age  1016.0  24.984252  5.080218 16.0   21.0   24.0   29.0   36.0
2      group 1016.0   3.011811  1.273395  1.0    2.0    4.0    4.0    4.0
3  asd_risk 1016.0   1.065945  0.248308  1.0    1.0    1.0    1.0    2.0
4  sum_qchat 1016.0  32.265748 12.136670  0.0   24.0   31.0   40.0   80.0
5      Q01  1016.0   1.021654  0.888639  0.0    0.0    1.0    1.0    4.0
6      Q02  1016.0   0.856299  0.820321  0.0    0.0    1.0    1.0    4.0
7      Q03  1016.0   1.898622  1.350992  0.0    1.0    2.0    3.0    4.0
8      Q04  1016.0   1.640748  1.382463  0.0    1.0    1.0    3.0    4.0
9      Q05  1016.0   1.083661  1.273407  0.0    0.0    1.0    2.0    4.0
10     Q06  1016.0   1.360236  1.358119  0.0    0.0    1.0    2.0    4.0
11     Q07  1016.0   1.180118  1.116839  0.0    0.0    1.0    2.0    4.0
12     Q08  1016.0   1.536417  1.172195  0.0    1.0    1.0    2.0    4.0
13     Q09  1016.0   1.515748  1.269478  0.0    1.0    1.0    2.0    4.0
14     Q10  1016.0   1.165354  1.199434  0.0    0.0    1.0    2.0    4.0
15     Q11  1016.0   1.156496  1.205127  0.0    0.0    1.0    2.0    4.0
16     Q12  1016.0   1.630906  1.287526  0.0    1.0    1.5    3.0    4.0
17     Q13  1016.0   1.292323  1.064510  0.0    0.0    1.0    2.0    4.0
18     Q14  1016.0   0.930118  0.903217  0.0    0.0    1.0    1.0    4.0
19     Q15  1016.0   1.929134  1.298312  0.0    1.0    2.0    3.0    4.0
20     Q16  1016.0   1.662402  1.356322  0.0    0.0    2.0    3.0    4.0
21     Q17  1016.0   1.395669  1.426333  0.0    0.0    1.0    3.0    4.0
22     Q18  1016.0   1.500984  1.307048  0.0    0.0    1.0    2.0    4.0
23     Q19  1016.0   1.215551  1.313737  0.0    0.0    1.0    2.0    4.0
24     Q20  1016.0   0.987205  1.316374  0.0    0.0    0.0    2.0    4.0
25     Q21  1016.0   1.105315  1.128114  0.0    0.0    1.0    2.0    4.0
26     Q22  1016.0   1.186024  1.031447  0.0    0.0    1.0    2.0    4.0
27     Q23  1016.0   1.108268  1.253209  0.0    0.0    1.0    2.0    4.0
28     Q24  1016.0   1.136811  1.031515  0.0    0.0    1.0    2.0    4.0
29     Q25  1016.0   0.769685  1.059838  0.0    0.0    0.0    1.0    4.0
30  Cluster 1016.0   0.689961  0.462737  0.0    0.0    1.0    1.0    1.0

Descriptive statistics for non-numeric variables:
  Variable Count Unique Top Freq
0  child_ID  1016    1016    288     1

```

Figure 3: Screenshot- Descriptive Statistics of Q-Chat Dataset Features

3.6.4 Data Characteristics

The dataset provides detailed insights into various behavioral and developmental features associated with autism spectrum disorder (ASD) in toddlers. The statistical summary reveals diverse scales and distributions across these features, indicating the need for normalization to prevent models from being biased toward variables with larger magnitudes. The dataset includes a mix of demographic variables such as sex and age, group classifications, and scores from a comprehensive questionnaire (Q-Chat) assessing various behavioral aspects. This rich data source supports the development of predictive models to identify ASD risk with precision and reliability.

3.6.5 Visualization of Target Variable and Features Distribution

The EDA began with bar charts to illustrate the frequency distribution of each ASD risk score, offering a clear view of the target variable's distribution. Following this, histograms and violin plots were used to show the distributions and densities of the features, revealing patterns and variations in the data.

Figure 4 illustrating the distribution of Autism Spectrum Disorder (ASD) risk scores shows that most individuals fall into the low-risk category. Specifically, 949 of the population is classified as having a low risk of ASD. In contrast, the high-risk category, which indicates a greater likelihood of ASD, covers a smaller proportion of the population, making up 67. This distribution highlights that most individuals in this population are considered to have a low risk of ASD based on their scores.

Figure 5 shows the distribution of each participant's score. Each participant scored each of the 25 items on a scale from 0 to 4, with higher scores indicating a typical behavior symptomatic of ASD. The KDE overlay offers a smoothed estimate of the probability density function, illustrating the spread and central tendency of the feature values.

The violin plots in Figure 6 reveal how different features (Q01 to Q25) are distributed across two ASD risk groups. Key insights include that Q01, Q08, and Q18 features show distinct patterns between the groups, indicating varying behaviors or responses. Some features, like Q04 and Q15, exhibit skewness, which might affect model accuracy, while others, like Q10 and Q21, have overlapping distributions, suggesting they may be less useful for classification. Features with clear separation, such as Q08 and Q22, are more valuable for predicting ASD risk. Uniform features like Q17 and Q19 provide consistent information across groups but might not be strong discriminators.

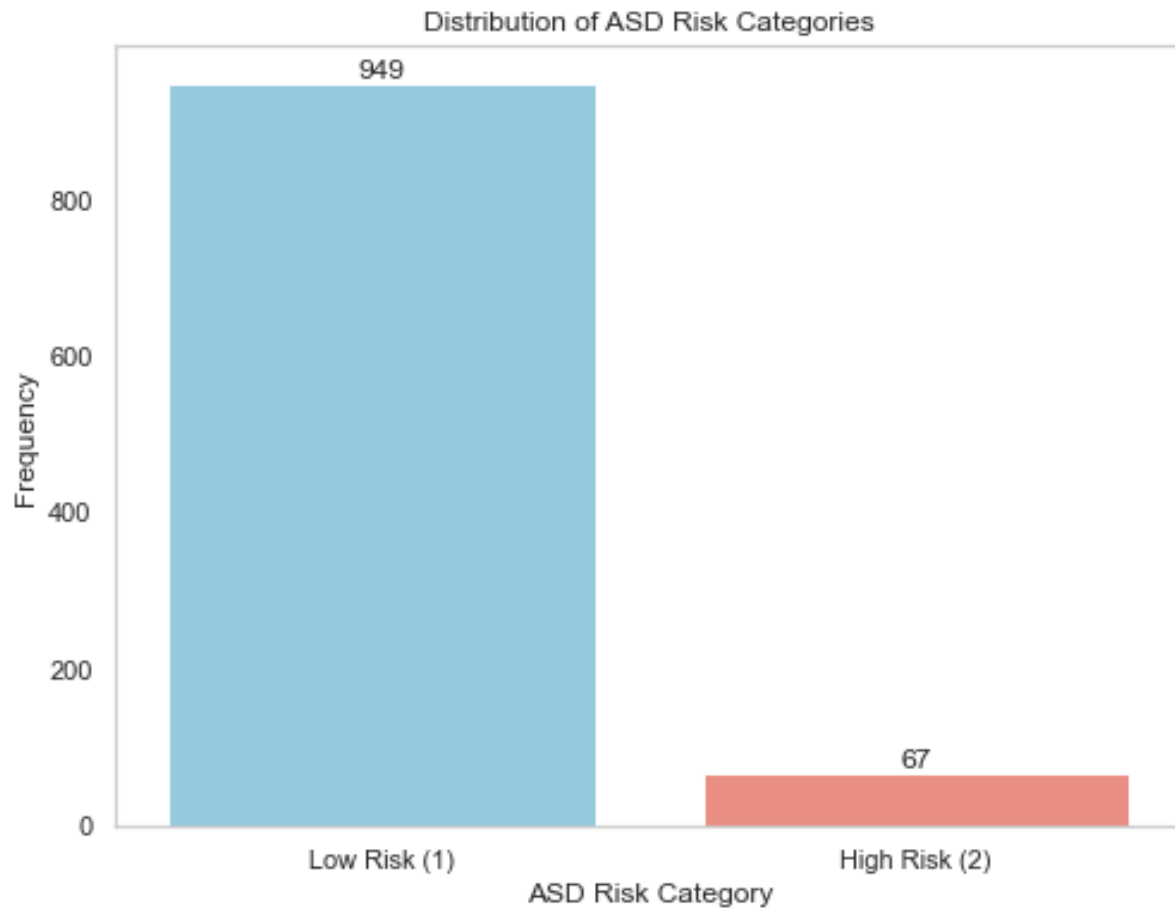


Figure 4: Bar Chart showing Distribution of Target Variable -ASD Risk Score

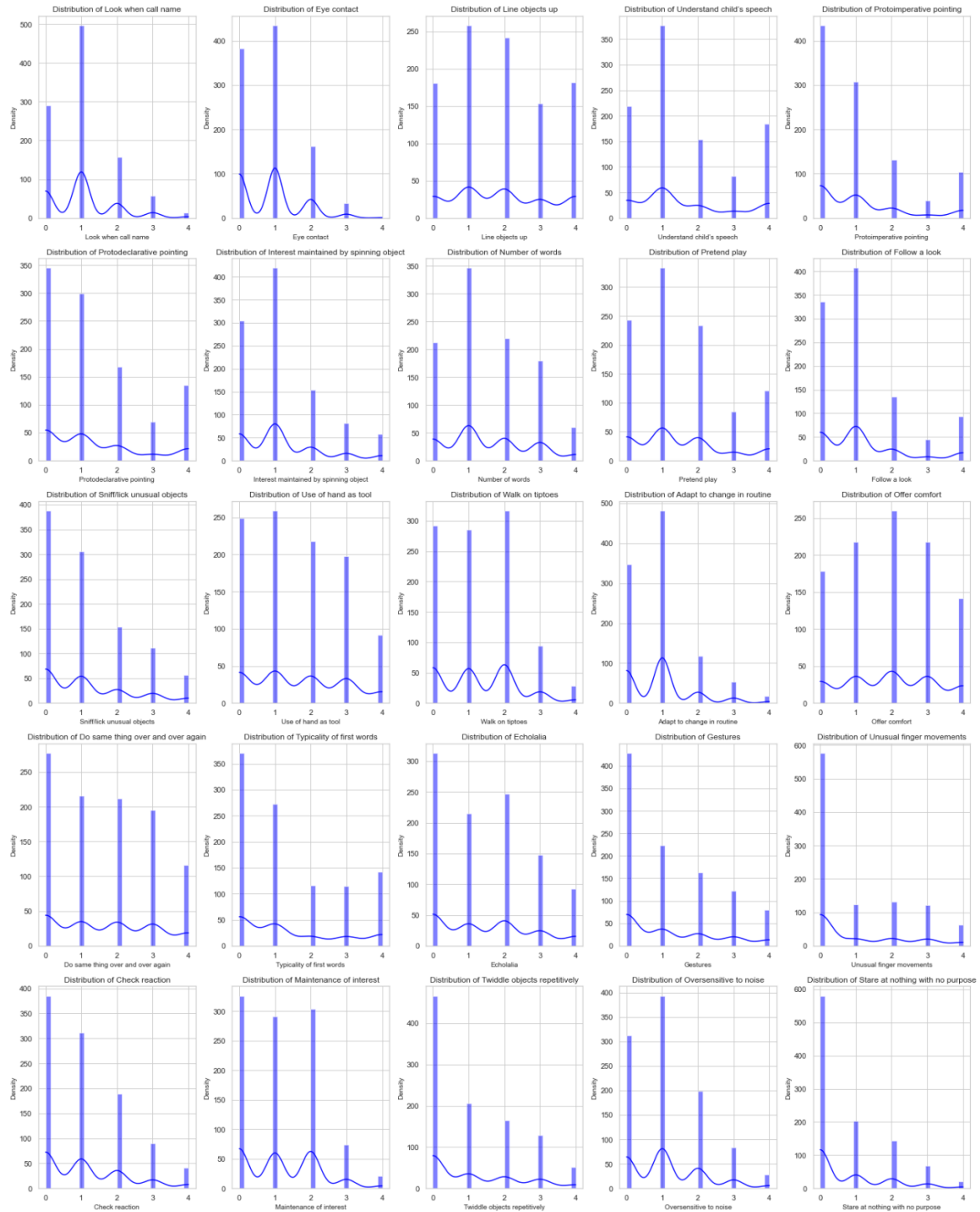


Figure 5: Comparative Distribution of 25 Questions by Responses through Histogram

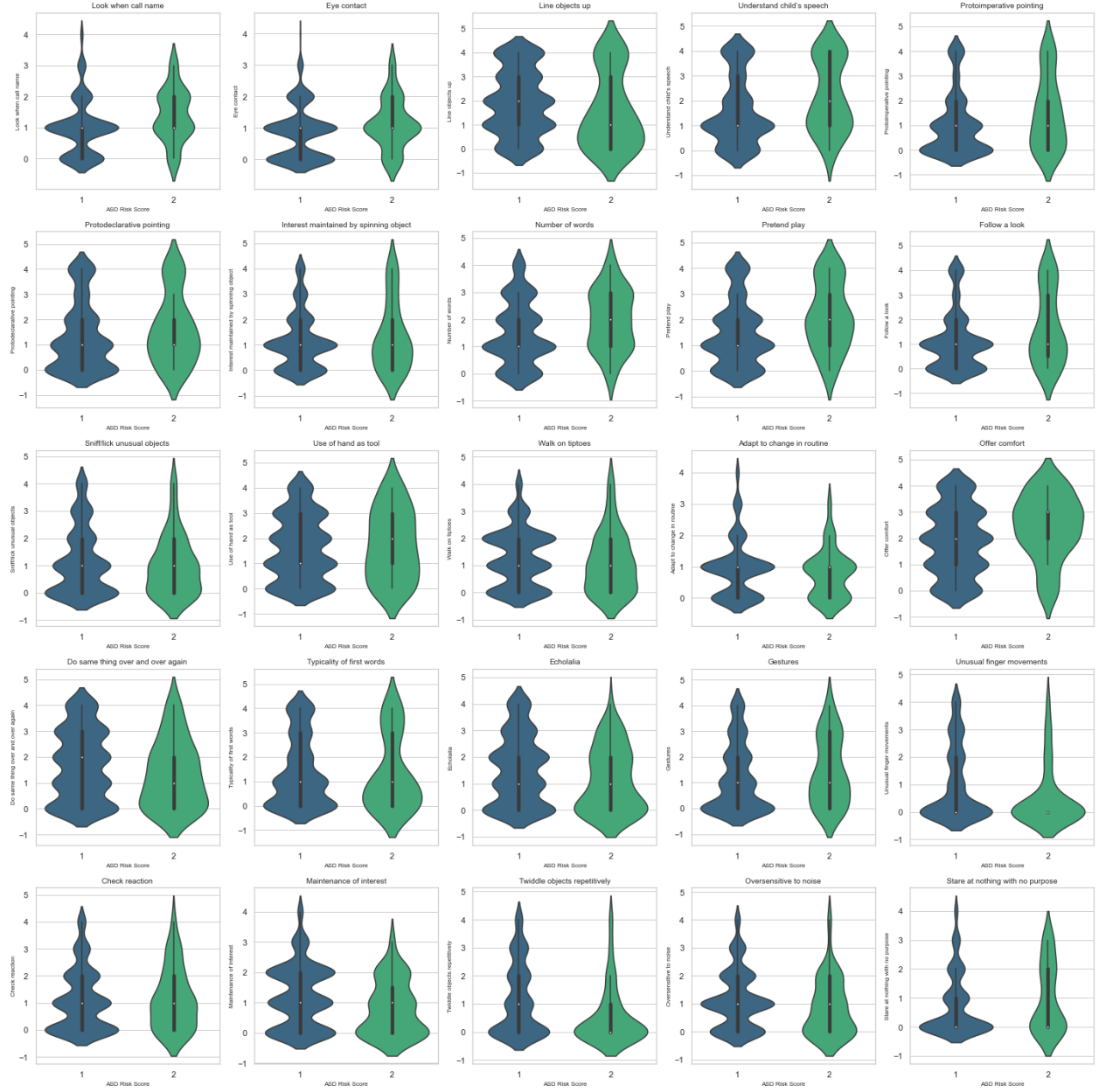


Figure 6: Violin Plot Showing Feature Distribution by ASD Risk Score

3.6.6 Correlation Analysis

The correlation matrix reveals linear relationships between features, with values ranging from -1 to 1, where positive correlations indicate that features increase together, and negative correlations show an inverse relationship. Strong positive correlations are observed between features like Q01, Q02, Q05, and Q06, while strong negative correlations exist between pairs such as Q08 and Q03. Moderate correlations among features suggest varying degrees of association, but the correlation

between the target variable, 'asd_risk', and individual features is weak, indicating that 'asd_risk' is influenced by multiple factors.

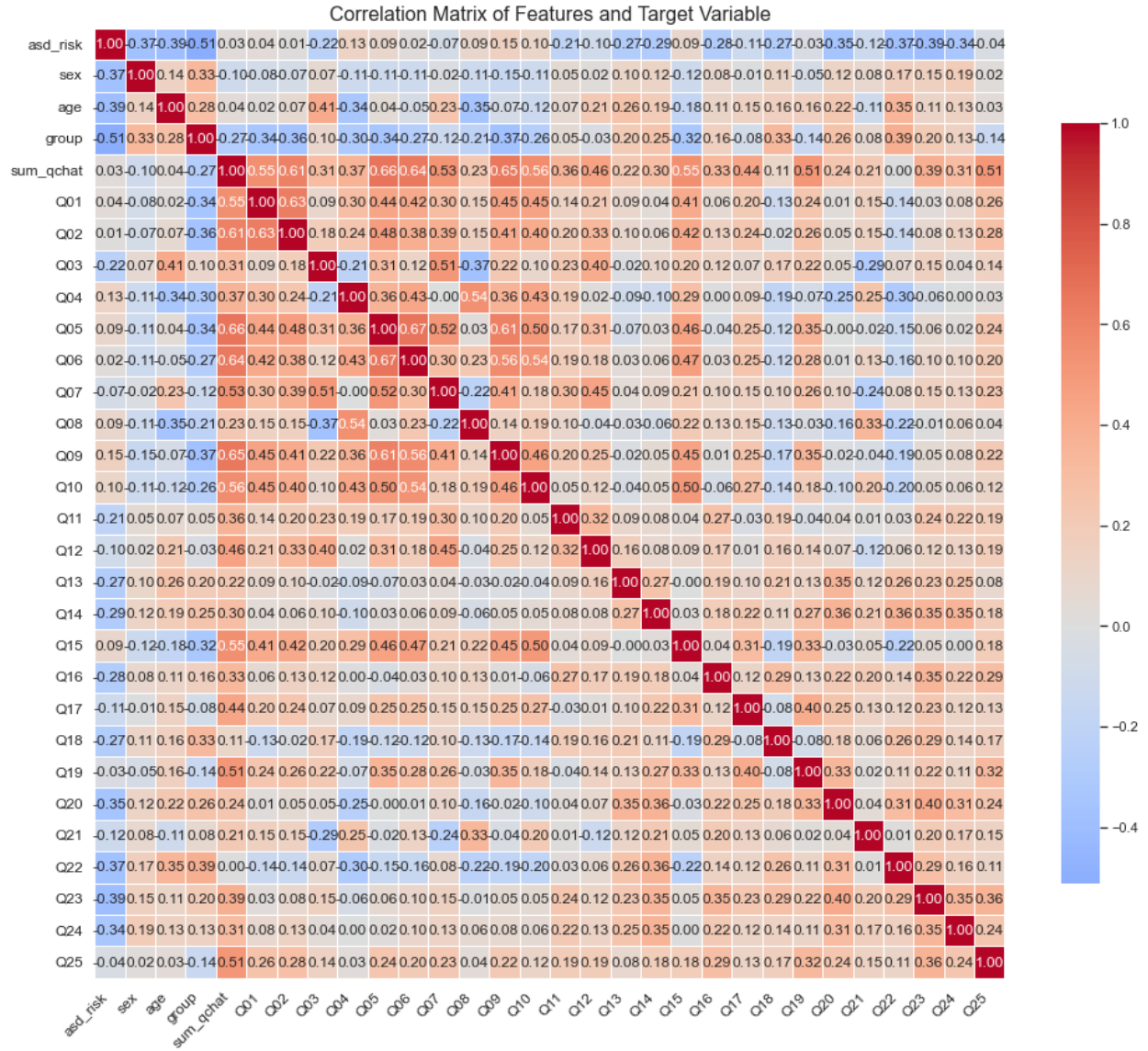


Figure 7: Heatmap of Feature Correlation in the Q-Chat Dataset

3.7 Data Preprocessing

Data preprocessing is a vital step in preparing the dataset for analysis and modeling. In this section, we undertook several key activities to ensure the quality and usability of the data:

3.7.1 Data Cleaning

The Q-Chat dataset underwent a comprehensive data-cleaning procedure to guarantee its quality for analysis. Examination revealed that it contained no missing values, eliminating the need for

imputation methods. This completeness of data enhances the reliability of the subsequent machine-learning models.

3.7.2 Data Integrity Validation

The dataset was thoroughly validated and found to have no missing values, ensuring its completeness and reliability. It includes varied data types and accurate identifiers. The sex column predominantly represents one gender, while ages range from 16 to 36 months. The group and asd_risk columns show balanced categories and low-risk predominance. The sum_qchat scores and individual question responses indicate a wide range of values, confirming the dataset's readiness for further analysis.

3.7.3 Categorical Variable Encoding

To prepare the dataset for machine learning algorithms, the target variable, asd_risk, was remapped from its original labels to binary values for clearer classification. Specifically, the original labels were transformed: 'Low Risk' (1) was mapped to 0, and 'High Risk' (2) was mapped to 1. This binary encoding simplifies the target variable and aligns it with the requirements of various machine-learning models. The dataset was then structured by dividing it into independent variables (features, X) and the dependent variable (target, y). In this setup, 'X' encompasses all the columns, while 'y' is composed exclusively of the 'asd_risk' column. This separation is essential for properly training the model.

3.7.4 Normalization/Feature Scaling

Normalization was performed using the Min-Max Scaling technique. This process scales the feature values to a range between 0 and 1, ensuring that each feature contributes equally to the model. For instance, features like Q01, Q02, etc., were transformed so that their values are bounded within this range, facilitating better performance of algorithms sensitive to the scale of input data.

Standardization was also applied, which adjusts the features to have a mean of 0 and a standard deviation of 1. This scaling technique helps to achieve a uniform distribution across features, making them comparable and improving the convergence rate of algorithms that use gradient descent. For example, features were transformed to have standardized values like those around a mean of 0 with a variance of 1.

By applying normalization and standardization, the dataset was transformed to ensure that all features are on a comparable scale, which improves the efficiency and effectiveness of machine-learning models. After normalizing and scaling, the data was transformed back into a Pandas DataFrame format to preserve its tabular structure.

3.8 Rasch Analysis

Rasch analysis was used to evaluate the difficulty levels of items in the Q-CHAT dataset. The analysis provided difficulty estimates for each question, ranging from -7.5629 (most challenging) to -1.8557 (least challenging). For example, Q02 was identified as the most difficult, while Q03 was the easiest. These difficulty levels were visualized in a bar chart, highlighting how each item varies in difficulty. This analysis aids in refining the questionnaire, enhancing its validity and reliability for detecting Autism Spectrum Disorder.

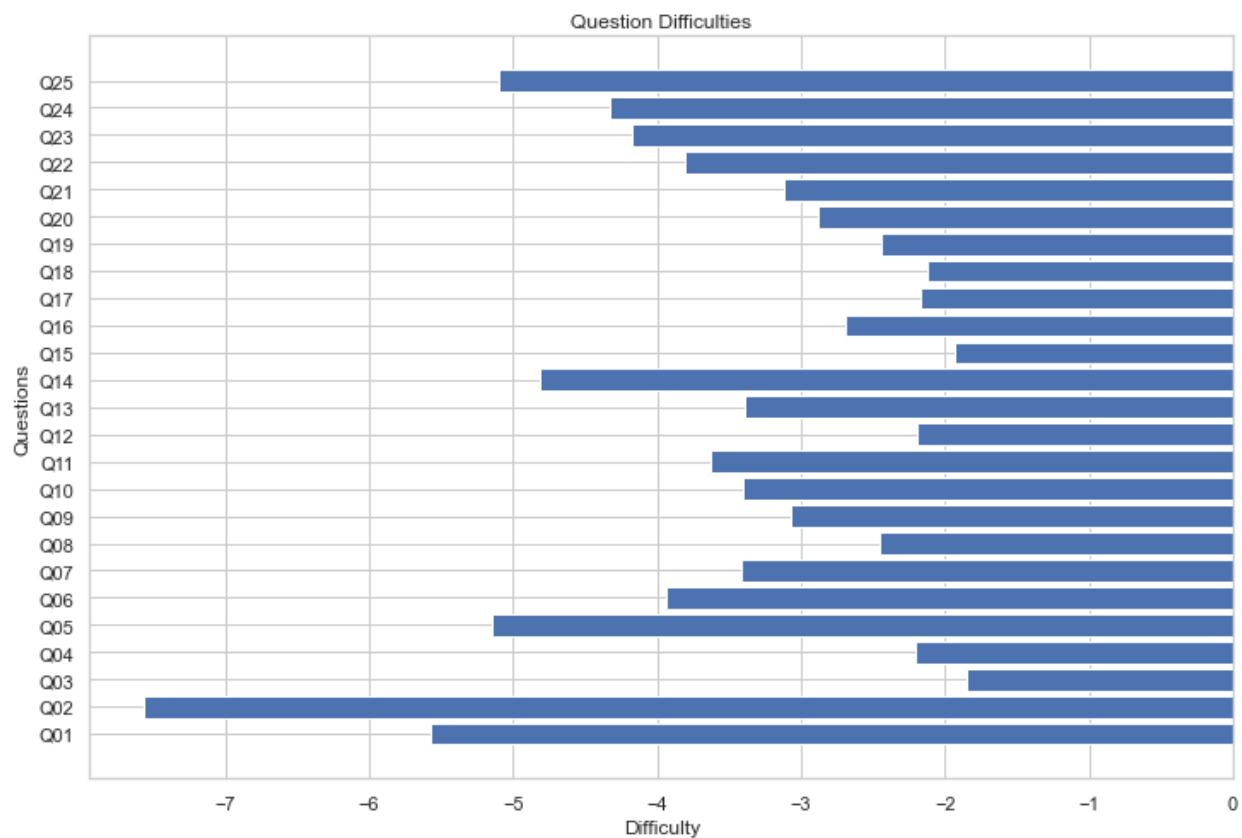


Figure 8: Question Difficulty Distribution

3.9 K-Means clustering

In K-Means clustering, the features were first standardized to ensure each contributed equally. The Elbow Method was used to determine the optimal number of clusters by plotting inertia. After deciding on the number of clusters, K-Means was applied, and the resulting clusters were added to the dataset. The Silhouette Score was calculated to evaluate the clustering quality, indicating that there might be room for improvement.

3.10 Principal Components Analysis (PCA)

In PCA, features were reduced to two dimensions for visualization. The principal components were added to the dataset, and clusters were plotted in this reduced space. A heatmap was created to show how different features contributed to each cluster. Principal component loadings were examined to understand the importance of each feature, revealing which features had the most influence on the principal components.

3.10 Model Training, Prediction, and Evaluation Framework

This section provides an overview of the systematic approach used for training, predicting, and evaluating machine learning models. The framework is designed to ensure consistency and reproducibility across different models and datasets.

3.10.1 Feature Selection

Feature selection is a critical step in model development, as it involves identifying the most relevant variables that contribute to the prediction task. In this framework, the selected features include Sex ('sex'), Age ('age'), Group ('group'), Sum of Q-CHAT Scores ('sum_qchat'), Individual Q-CHAT Questions ('Q01' to 'Q25'). These features serve as the input variables ('X'), with the target variable ('y') being the binary classification label 'asd_risk', indicating the risk of Autism Spectrum Disorder (ASD).

3.10.2 Data Splitting

The dataset is divided into training and testing subsets to evaluate the model's performance on unseen data. The data is split with the training set being 80% of the data, used to train the models with the testing set being 20% of the data, reserved for model evaluation. A stratified split is applied to ensure the class distribution remains consistent across the training and testing sets.

3.10.3 Handling Class Imbalance

Class imbalance was seen in the target variable with class 1 (high risk score) being underrepresented. To address this, Synthetic Minority Over-sampling Technique (SMOTE) was

applied to the target variable before proceeding to machine learning. SMOTE generates synthetic samples for the minority class, balancing the class distribution and enabling the model to learn more effectively from both classes.

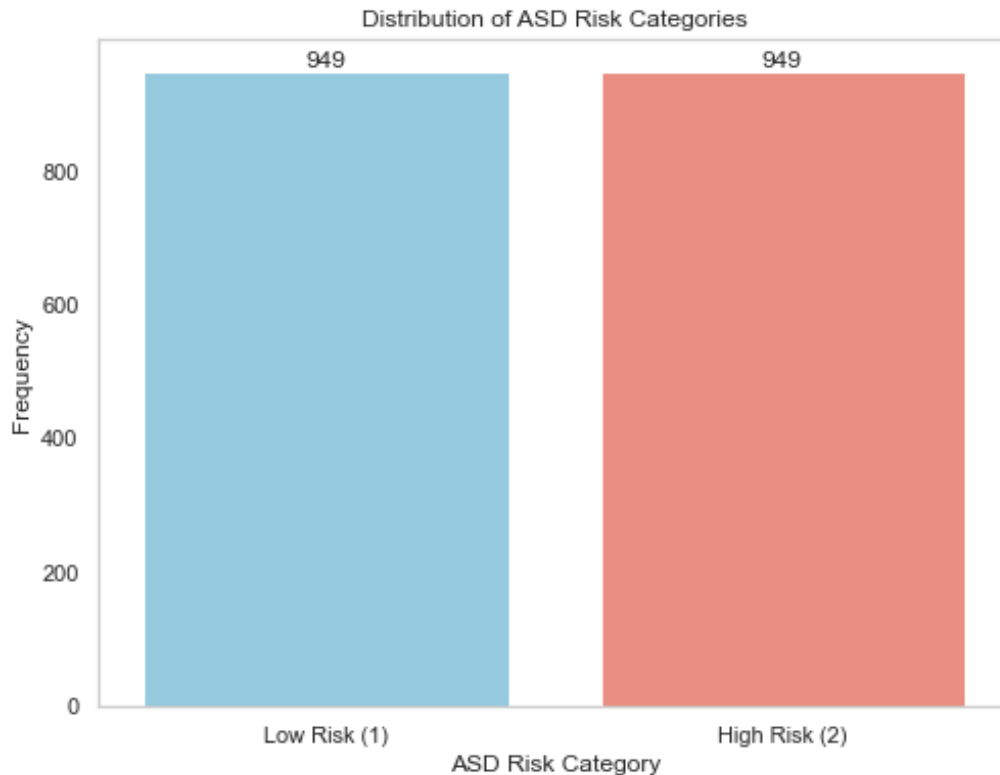


Figure 9: Distribution of ASD Risk Categories after SMOTE

3.10.4 Model Initialization

Six different machine learning models are initialized to provide a diverse set of algorithms for comparison: Logistic Regression, K-Nearest Neighbor, Random Forest, Support Vector Machine (SVM), Decision Tree, and Gradient Boosting. Each model offers unique advantages and challenges, allowing for a comprehensive evaluation of their performance on the given task.

3.10 Evaluation Criteria

Key evaluation metrics were selected for their relevance to the classification task, to assess the performance of the predictive models in this research. The Q-chat dataset includes features from behavioral assessments and questionnaires designed to identify individuals at risk of Autism Spectrum Disorder (ASD). Accurate ASD risk prediction is essential for early intervention, choosing robust evaluation metrics is critical. These metrics provide a comprehensive view of model performance, ensuring accurate and reliable predictions.

3.10.1 Accuracy

Measures the proportion of correct predictions (both true positives and true negatives) out of all predictions.

$$\text{Mathematically, Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Where:

TP (True Positives) is the number of positive samples correctly identified by the model.

TN (True Negatives) is the number of negative samples correctly identified by the model.

FP (False Positives) is the number of negative samples incorrectly identified as positive.

FN (False Negatives) is the number of positive samples incorrectly identified as negative.

3.10.2 Precision

Precision measures the proportion of true positive predictions out of all positive predictions.

Mathematically:

$$\text{Precision} = \frac{TP}{TP+FP}$$

3.10.3 Recall

Recall measures the proportion of positive instances correctly identified by the model.

Mathematically:

$$\text{Recall} = \frac{TP}{TP+FN}$$

3.10.4 F1 Score

Balances precision and recall, providing a single metric to assess a model's performance.

$$\text{F1 Score} = \frac{2 \times (\text{precision} \times \text{Recall})}{\text{Recall} + \text{Precision}}$$

3.10.5 AUC-ROC (Area Under the ROC Curve)

The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate (FPR). The AUC-ROC score represents the area under this curve, providing an aggregate measure of performance across all classification thresholds.

3.10.5.1 True Positive Rate (TPR) or Recall:

$$\text{TPR} = \frac{TP}{TP + FN}$$

3.10.5.2 False Positive Rate (FPR):

$$FPR = \frac{FP}{FP + TN}$$

3.10.5.3 AUC-ROC is typically computed using numerical integration of the ROC curve:

$$AUC - ROC = \int_0^1 TPR(FPR)d(FPR)$$

3.11 Visualization of Model Performance

Visual tools such as the confusion matrix and ROC curve are employed to better understand model performance:

3.11.1 Confusion Matrix

A Confusion Matrix is a visual tool used to evaluate the performance of classification algorithms. It organizes prediction results into a grid where columns represent predicted classes and rows represent actual classes. In a binary classification scenario, the matrix has dimensions 2×2, showing the counts of true positives, false positives, true negatives, and false negatives. This matrix helps in calculating various performance metrics, such as precision and recall, which are applicable to all classification algorithms (Xu, Zhang & Miao, 2020).

3.11.2 ROC Curve and AUC Score

ROC curves are used to evaluate the performance of a predictor by plotting the true positive rate against the false positive rate across all thresholds. This analysis is threshold-agnostic, allowing for the selection of an optimal threshold based on specific criteria. The model's overall performance is summarized by the Area Under the Curve (AUC), which quantifies its discriminative ability (Muschelli, 2020).

3.12 Cross-Validation

The study employs 5-fold cross-validation to evaluate the performance of machine learning models on the Q-CHAT dataset. The dataset is divided into five equal parts or folds. Each fold is used as a test set once, while the remaining four folds are combined to form the training set. This process is repeated five times, ensuring every data point is used for training and testing. Stratified sampling is applied to maintain class distribution, enhancing the reliability and generalizability of the model evaluations.

3.13 Model Interpretability- LIME (Local Interpretable Model-agnostic Explanations)

Analysis for Model Predictions

LIME (Local Interpretable Model-agnostic Explanations) was used to enhance the interpretability of the predictive models by explaining individual predictions. For each model, Logistic Regression, K-Nearest Neighbor, Random Forest, SVM, Decision Tree, Gradient Boosting, and Stacking Classifier, LIME provided insights into how features influenced specific predictions. By approximating complex models with simpler, interpretable ones around individual instances, LIME generated visual explanations which highlighted feature contributions thus improving transparency and trust in the model's decisions.

4. Chapter Four: Results, Analysis and Discussion

This research meticulously examines various machine learning models which include Logistic Regression, Random Forest, Support Vector Machine, K-Nearest Neighbor, Decision Tree, Gradient Boosting and Stacking method to determine their effectiveness in detecting Autism Spectrum Disorder (ASD) in toddlers from Q-CHAT data. The analysis aims to identify the most accurate models for early ASD detection, assess the significance of various predictive features through feature importance analysis, and understand each model's performance in the context of ASD diagnosis and intervention. Additionally, the study employs LIME (Local Interpretable Model-agnostic Explanations) to improve the interpretability of these classifiers, making their predictions more transparent and actionable for clinicians.

The models were evaluated on their ability to classify autism risk in toddlers using accuracy, precision, recall, F1 score, and ROC AUC. The evaluation was conducted using a test set split from the balanced dataset and metrics were calculated as follows:

- a) **Accuracy** reflects the proportion of correctly classified instances.
- b) **Precision** measures the proportion of true positive predictions among all positive predictions.
- c) **Recall** represents the model's ability to correctly identify actual positive cases.
- d) **F1 Score** provides a balance between precision and recall.
- e) **ROC AUC** assesses the model's ability to distinguish between classes.

4.1 Baseline Model Performance with Complete Feature Set

Establishing a baseline model performance using the complete feature set from the Q-CHAT dataset is a critical first step in our analysis of detection of ASD in toddlers. This initial evaluation serves as a foundation for our subsequent investigations into more advanced model configurations and feature reduction techniques. The importance of this baseline analysis can be summarized as follows:

Performance Benchmark: This will provide a reference point against which we can compare the performance of the models using the selected feature set.

Model Behavior Understanding: This will help us understand how different machine learning algorithms behave with the complete dataset, revealing their strengths and limitations.

Feature Importance Insight: Using the full feature set initially allows us to gain a comprehensive understanding of which features contribute most significantly to detection of ASD.

4.1.1 Comparative Evaluation of Model Performance Using the Complete Feature Set

The comparative evaluation of model performance shown in Figure 10, using the complete feature set from the Q-CHAT dataset reveals significant insights into the effectiveness of various machine learning algorithms for autism detection in toddlers.

Support Vector Machine (SVM) demonstrates exceptional performance with an accuracy of 96.8% and perfect recall of 1.0. It achieves a high precision of 0.941 and an F1 score of 0.969, reflecting a robust balance between precision and recall. The SVM model also exhibited the highest ROC AUC score of 0.996, indicating its superior ability to distinguish between autistic and non-autistic cases.

Random Forest follows closely with an accuracy of 96.3%, and it maintains a strong balance with a precision of 0.958 and recall of 0.968. Its F1 score of 0.963 and ROC AUC score of 0.990 further underscore its effective classification capabilities and robust performance across different thresholds.

Gradient Boosting shows strong overall performance with an accuracy of 93.9% and high recall of 0.974. It has a precision of 0.911 and an F1 score of 0.941. Its ROC AUC of 0.977 highlights its excellent discriminative power in detecting autism.

K-Nearest Neighbors (KNN) achieves a perfect recall of 1.0 but has a lower precision of 0.815, resulting in an accuracy of 88.7%. While its ROC AUC score of 0.963 indicates good performance, it does not match the top models in terms of precision and overall accuracy.

Decision Tree provides respectable results with an accuracy of 90.3% and a recall of 0.958. It has a precision of 0.863 and an F1 score of 0.908. However, its ROC AUC score of 0.903 suggests that it may not discriminate between autistic and non-autistic cases as effectively as the leading models.

Logistic Regression shows an accuracy of 85.8% and a good recall of 0.916. With a precision of 0.821 and an F1 score of 0.866, it has the lowest accuracy among the models but maintains a reasonably high ROC AUC score of 0.925, indicating satisfactory discriminative ability.

The evaluation reveals that SVM, Random Forest, and Gradient Boosting demonstrated the most promising results across all metrics for autism detection in toddlers, with SVM standing out as the top performer.

Model Performance Comparison (All Features)

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.858	0.821	0.916	0.866	0.925
SVM	0.968	0.941	1	0.969	0.996
KNN	0.887	0.815	1	0.898	0.963
Random Forest	0.963	0.958	0.968	0.963	0.99
Decision Tree	0.903	0.863	0.958	0.908	0.903
Gradient Boosting	0.939	0.911	0.974	0.941	0.977

Figure 10: Screenshot- Summary of Models' Performance on Full Features Set

We visually compare the performance of six machine learning models across various metrics, Accuracy, Precision, Recall, F1 Score, and ROC AUC in Figure 11. This format facilitates the identification of the most reliable and balanced models, thereby aiding in the selection of the optimal model for specific use cases.

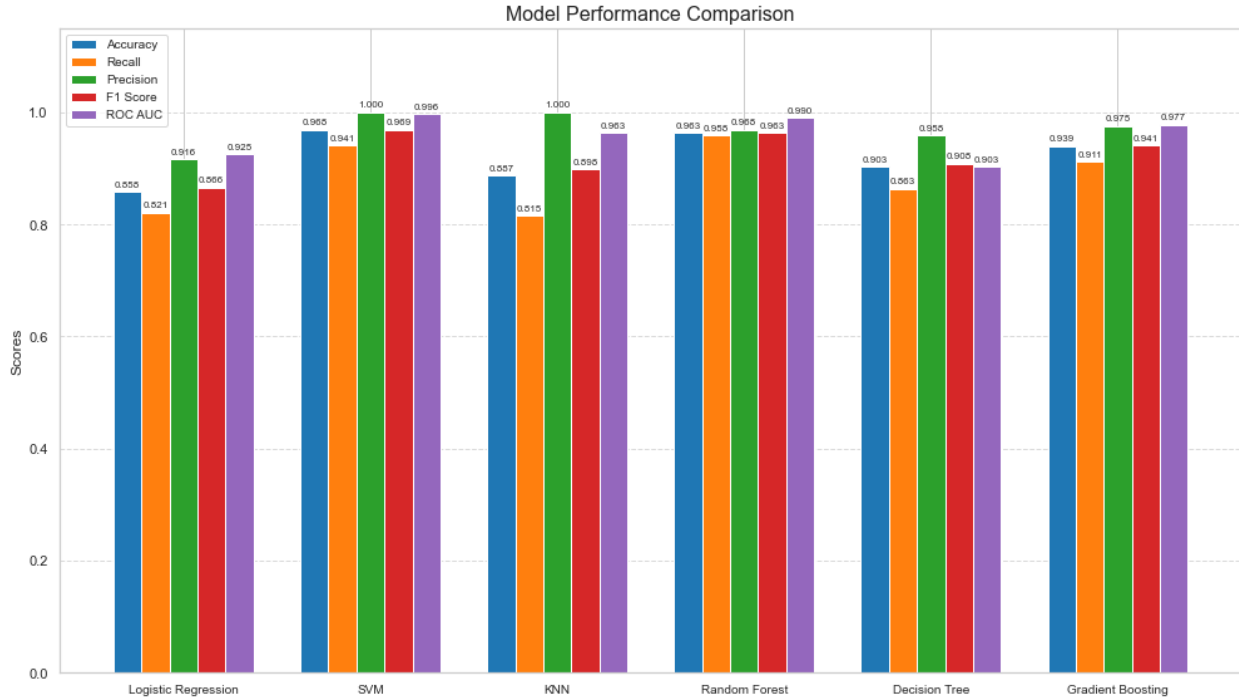


Figure 11: Representation of Machine Learning Model Efficacy Across Five Metrics

4.1.2 Comparative Presentation of Model Performance: Confusion Matrix Analysis

As illustrated in Figure 12, Logistic Regression demonstrates solid performance, with 174 true positives and 152 true negatives, indicating a reasonable balance between sensitivity and specificity, despite 38 false positives and 16 false negatives. SVM stands out with exceptional results, achieving 190 true positives and 178 true negatives, marked by high accuracy, zero false negatives, and only 12 false positives.

KNN also shows strong performance with 190 true positives and 147 true negatives, achieving perfect sensitivity. However, it has a higher false positive rate of 43 compared to other models. Random Forest offers robust results with 184 true positives and 182 true negatives, maintaining a favorable balance between sensitivity and specificity, with just 8 false positives and 6 false negatives.

The Decision Tree model exhibits moderate performance with 182 true positives and 161 true negatives, reflecting a higher false positive rate of 29 and a moderate false negative rate of 8. Gradient Boosting displays strong performance with 185 true positives and 172 true negatives, effectively balancing sensitivity and specificity, with 18 false positives and 5 false negatives.

These confusion matrices highlight each model's diagnostic accuracy, emphasizing the important trade-offs between sensitivity and specificity. This comparative analysis is crucial in selecting the optimal model for specific applications, underlining the importance of carefully evaluating the desired balance between correctly identifying both "Low Risk" and "High Risk" cases.

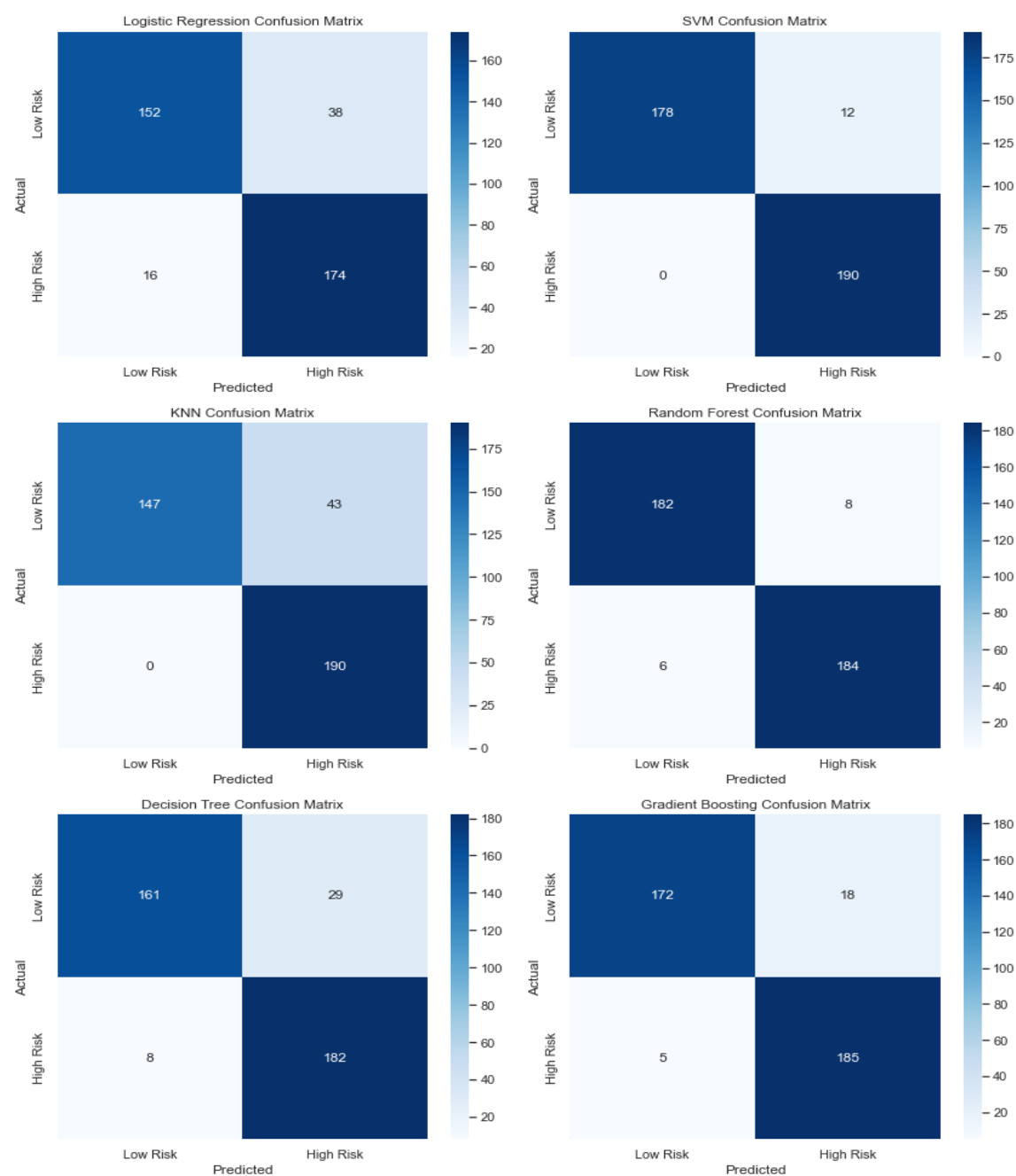


Figure 12: Performance of ML Classifiers in ASD Detection in Toddlers: Confusion Matrices Overview

4.1.3 ROC Curve Analysis Across Machine Learning Models

The ROC curves and AUC values for the six machine-learning models are illustrated in Figure 13. The SVM model stands out with an exceptional AUC of 0.996, showcasing its superior ability to differentiate between classes. Random Forest follows closely with an AUC of 0.990, indicating strong performance. Gradient Boosting also demonstrates robust classification with an AUC of 0.977. Logistic Regression shows a commendable AUC of 0.925. KNN, despite achieving perfect recall, has a lower AUC of 0.963, which suggests some limitations in its class discrimination across different thresholds. The Decision Tree model, with an AUC of 0.903, displays the least discriminatory capacity among the models. These results highlight the importance of the AUC metric in evaluating model performance, as it offers a comprehensive view of a model's effectiveness across various classification thresholds.

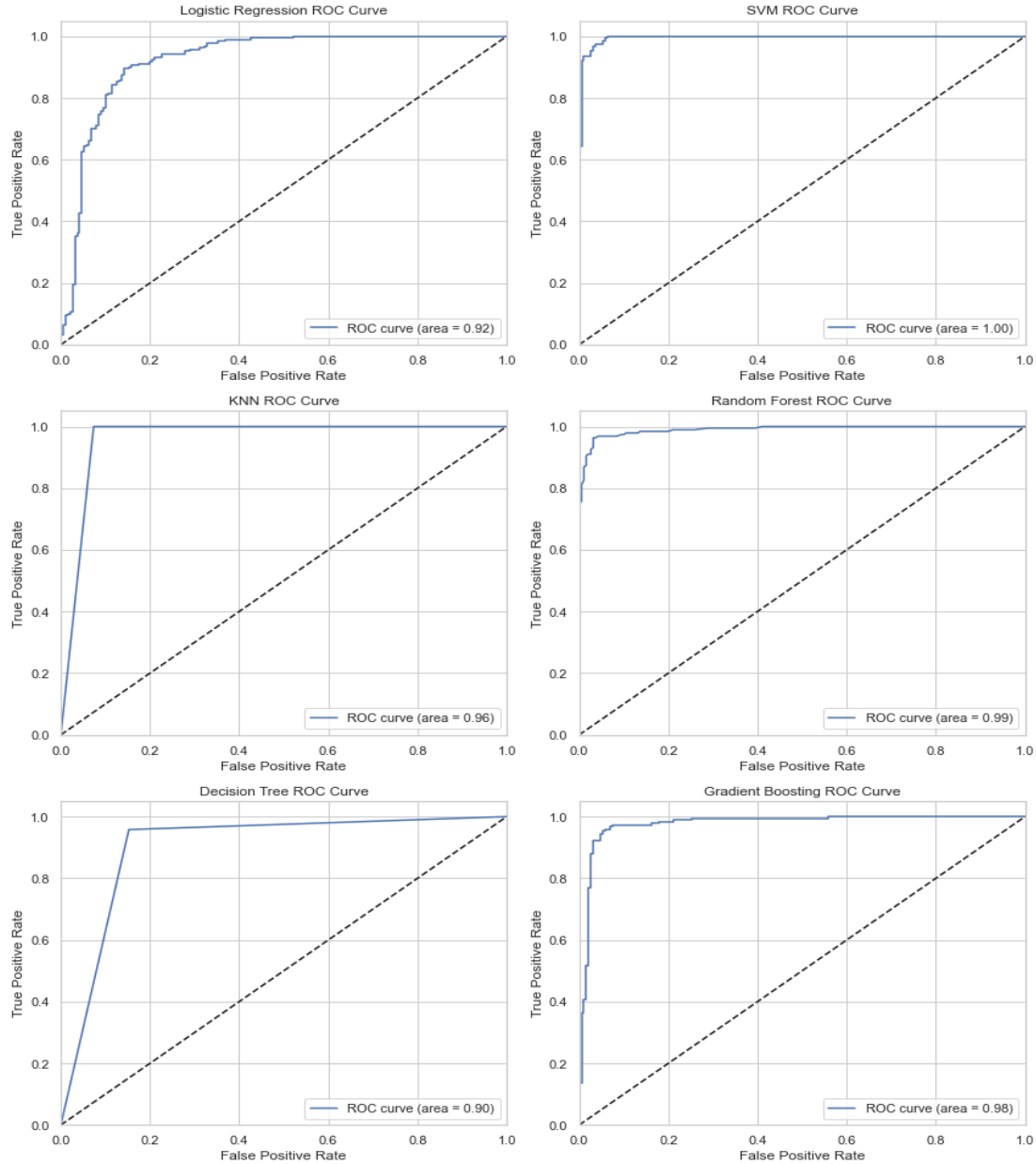


Figure 13: Evaluation of ROC Curves for Various Machine Learning Models

4.2 Comparison of Performance of Models After Feature Selection

In the building of predictive models for autism detection, precision and reliability are paramount. Recursive Feature Elimination (RFE) was utilized to refine the models by focusing on the most significant features. This section outlines the outcomes of applying RFE to Logistic Regression, SVM, Random Forest, Decision Tree, KNN, and gradient-boosting machine learning algorithms.

4.2.1 Summary of Model Performance After Feature Selection Using Recursive Feature Elimination (RFE)

The performance metrics of various models following the application of Recursive Feature Elimination (RFE) are shown in Figure 14. Among the evaluated models, SVM stands out as the top performer, achieving an impressive accuracy of 99.2%, a precision of 98.4%, and perfect recall. This results in an outstanding F1-score of 0.992, showcasing its exceptional classification accuracy.

Random Forest also demonstrates robust performance with an accuracy of 96.3% and balanced precision and recall, which resulted in an F1-score of 0.963. Similarly, Gradient Boosting provides solid results with an accuracy of 93.7% and a good equilibrium between precision and recall, resulting in an F1-score of 0.938.

KNN exhibits a high recall rate of 100%, yet it has a lower accuracy of 93.9% and precision of 89.2%, culminating in an F1-score of 0.943. Logistic Regression and Decision Tree show marked improvements after feature selection but still lag behind the top models in terms of overall accuracy and F1-score.

In conclusion, the RFE implementation has significantly enhanced the performance of all models. SVM, Random Forest, and Gradient Boosting emerge as the top performers, offering a commendable balance between accuracy, precision, recall, and F1-score.

Model Performance Comparison (Selected Features)						
Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	
Logistic Regression	0.842	0.793	0.926	0.854	0.853	
SVM	0.992	0.984	1	0.992	0.999	
KNN	0.939	0.892	1	0.943	0.984	
Random Forest	0.963	0.963	0.963	0.963	0.993	
Decision Tree	0.913	0.887	0.947	0.916	0.913	
Gradient Boosting	0.937	0.915	0.963	0.938	0.985	

Figure 14: Screenshot Comparison of the Models' Performance on Selected Features Set

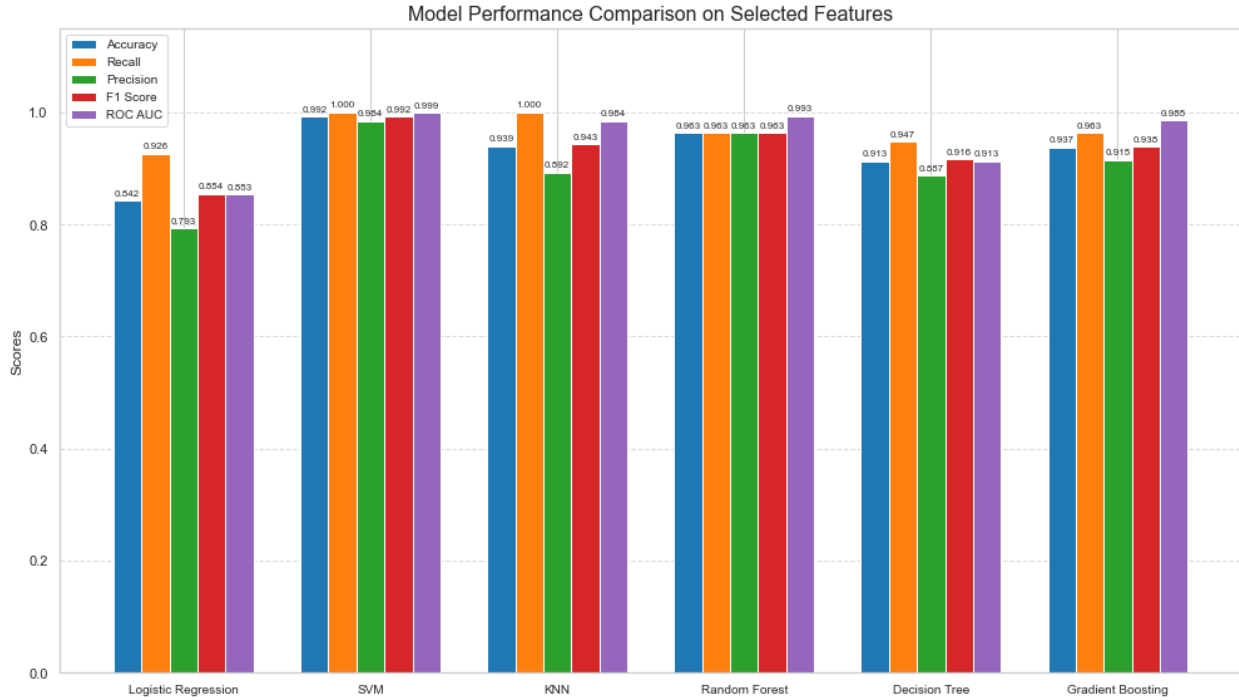


Figure 15: Visualization of Machine Learning Model Efficacy Across Five Metrics After Feature Selection

4.2.2 Comparative Analysis of Machine Learning Models Performance After Feature Selection

(a) ROC Curves and AUC Values for Machine Learning Models

After applying feature selection using Recursive Feature Elimination (RFE), the models were evaluated based on their ROC curves and AUC values. SVM emerged as the top performer with an exceptional AUC of 0.999, indicating its strong ability to distinguish between positive and negative classes. Random Forest and Gradient Boosting also showed excellent performance with high AUC values, demonstrating their effectiveness in classification tasks. KNN exhibited a good performance but was slightly behind the top-performing models. Logistic Regression and Decision Tree had lower AUC values, suggesting limitations in their classification capabilities. Overall, the models benefited from the feature selection process, with most showing improved performance compared to previous iterations (See Figure 16).

(b) Confusion Matrices for Machine Learning Models

After applying Recursive Feature Elimination (RFE), the performance of all models has notably improved, particularly in reducing false positives and false negatives.

SVM exhibits exceptional performance, achieving 190 True Positives and 187 True Negatives, with only 3 False Positives and no False Negatives, highlighting its near-perfect accuracy and minimal errors. KNN also performs strongly, matching SVM with 190 True Positives but with a slightly higher rate of 23 False Positives. Random Forest demonstrates balanced performance, recording 183 True Positives and 183 True Negatives, with moderate rates of 7 False Positives and 7 False Negatives. The Decision Tree shows a drop in accuracy, with 180 True Positives and 167 True Negatives, alongside 23 False Positives and 10 False Negatives. Logistic Regression has the lowest accuracy, with 176 True Positives, 144 True Negatives, 46 False Positives, and 14 False Negatives.

Overall, RFE has significantly enhanced the accuracy of the models, with SVM, Random Forest, and Gradient Boosting showing the most considerable improvements. SVM emerges as the top performer with near-perfect results post-feature selection (See Figure 17).

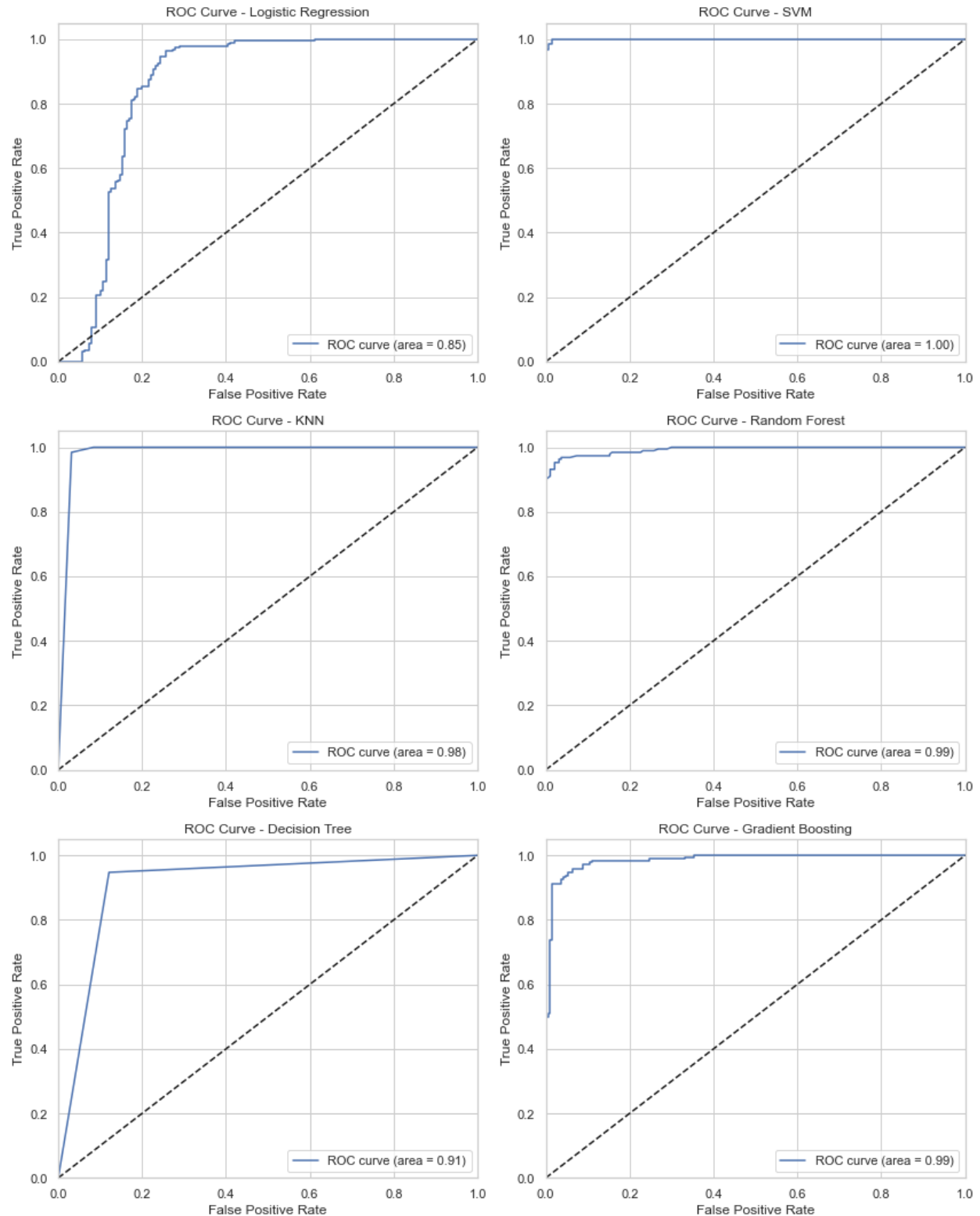


Figure 16: ROC Curves and AUC values for Machine Learning Models After Feature Selection

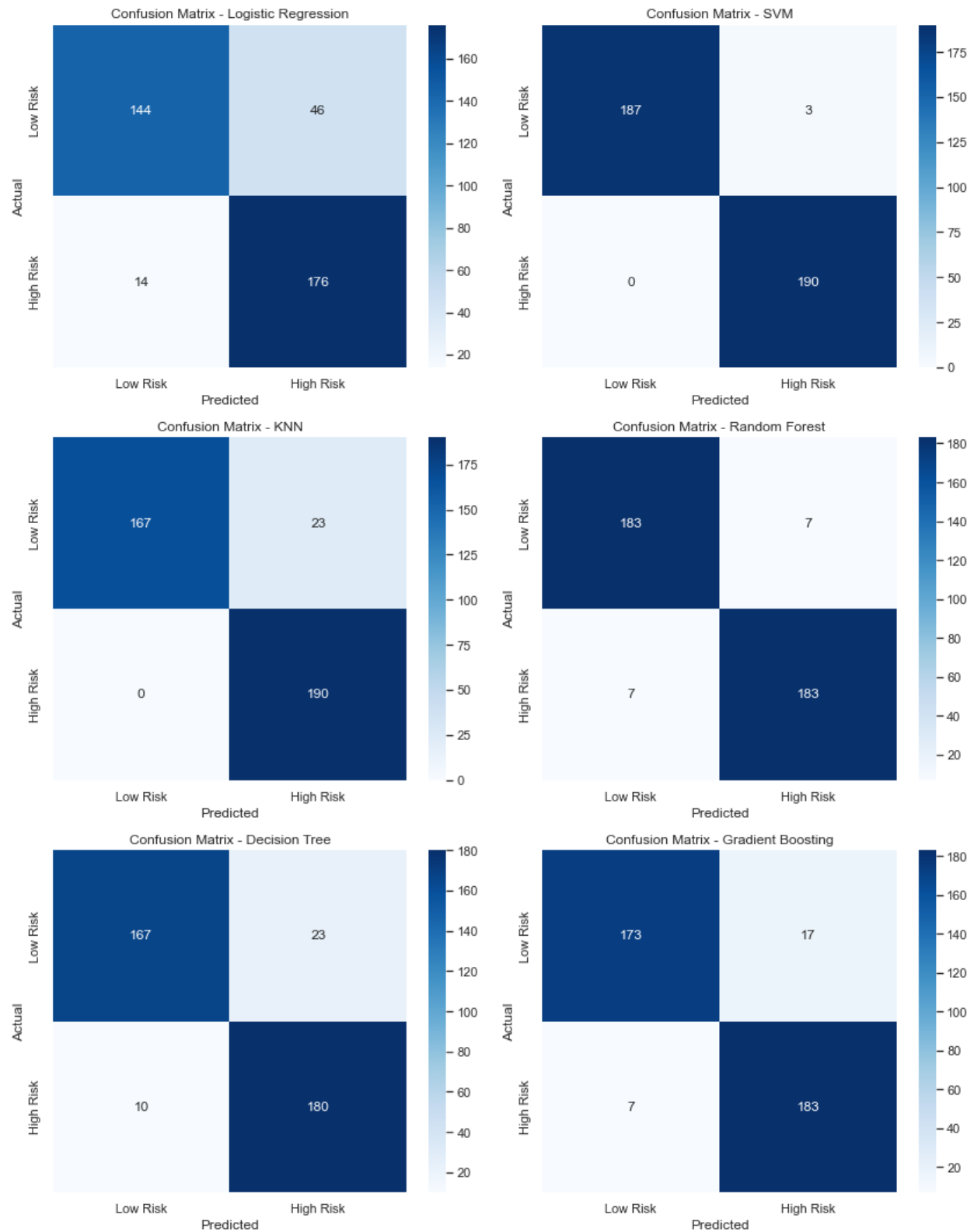


Figure 17: Confusion Matrices for Machine Learning Models After Feature Selection

4.3 Cross-Validation Performance Analysis with Selected Features

The cross-validation results confirmed the stability and effectiveness of the machine-learning models after feature selection. SVM emerged as the top performer, achieving an average accuracy of 96.0%, high precision (93.3%), perfect recall (100%), and an average F1 score of 96.2%. Its ROC AUC score of 99.4% further underscores its robustness in distinguishing between classes. Random Forest and Gradient Boosting also demonstrated strong performance, with average accuracies of 93.3% and 90.6%, respectively. Random Forest maintained high precision and recall, while Gradient Boosting showed good balance across metrics. Logistic Regression, KNN, and Decision Tree models exhibited average accuracies of 84.6%, 90.7%, and 90.6%, respectively. Despite being slightly less accurate, these models provided solid performance, particularly in their ability to detect autism risk. Overall, feature selection significantly enhanced the performance of all models, contributing to their strong and reliable results.

4.4 Summary of Stacking Classifier Performance

The stacking classifier, which integrates various base models with a Logistic Regression meta-classifier, demonstrated exceptional performance on the test set. It achieved an accuracy of 98.9%, precision of 99.5%, recall of 98.4%, F1 score of 98.9%, and a perfect ROC AUC of 1.000. Cross-validation results further confirmed its effectiveness with an average accuracy of 97.0%, precision of 96.4%, recall of 97.6%, F1 score of 97.0%, and an average ROC AUC of 99.7%. These metrics underscore the stacking model's robustness and high reliability in predicting autism risk across various data subsets.

4.5 Feature Importance Analysis

The feature importance analysis across Logistic Regression, Support Vector Machine (SVM), Random Forest Classifier (RFC), Gradient Boosting, and Decision Tree (DT) highlights the key features influencing predictions.

Key features consistently recognized across multiple models include Q24, Q23, group, and Q20. Logistic Regression and SVM emphasize Q24 and Q23, indicating their strong predictive value. Tree-based models like RFC, Gradient Boosting, and Decision Trees also highlight these features, reflecting their vital role in model decisions.

The uniform importance of these features across diverse models underscores their reliability and significance. This consistency enhances the models' predictive accuracy and provides a solid foundation for further development and real-world application (See Figures 17 and 18).



Figure 18: Feature Importance Analysis Across Machine Learning Models

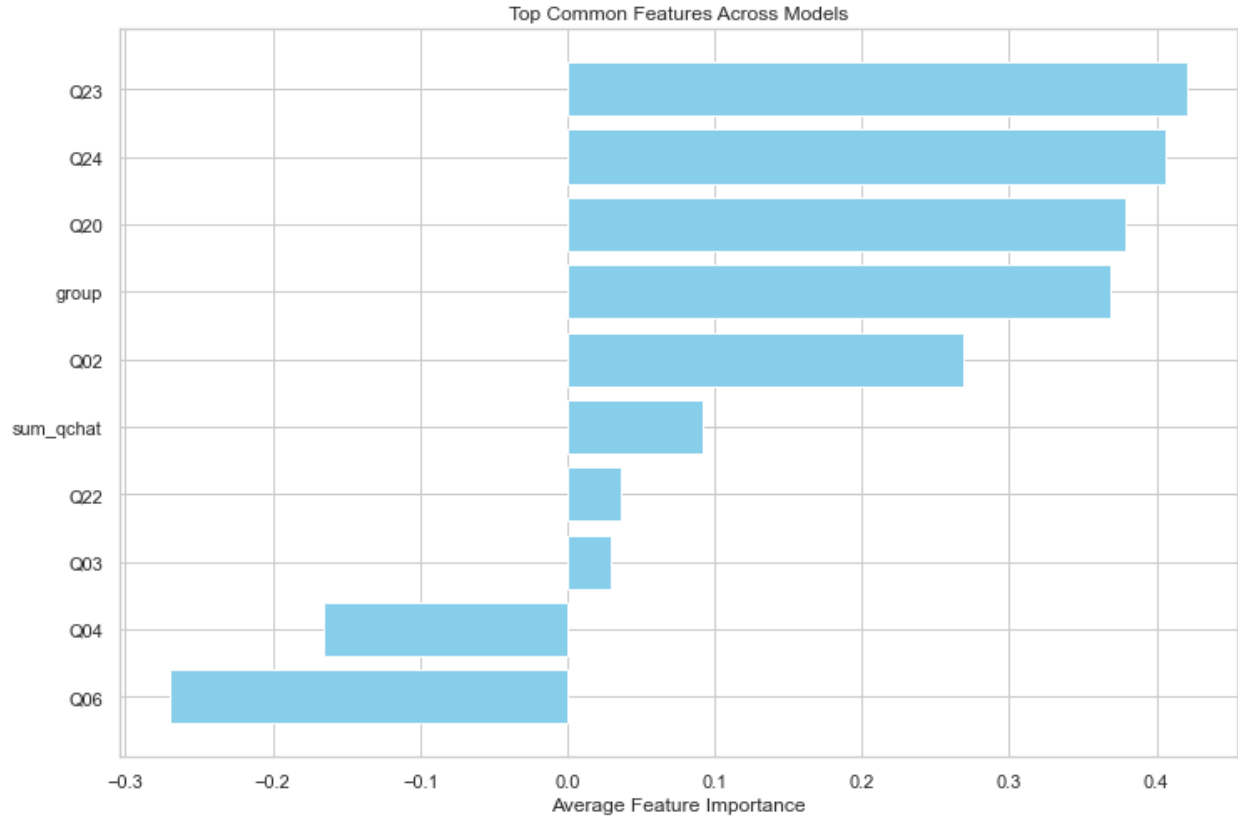


Figure 19: Average Feature Importance Across Multiple Machine Learning Models

4.6 Local Interpretable Model-agnostic Explanations (LIME) Analysis for the Models

Local Interpretable Model-agnostic Explanations (LIME) analysis provides insight into the contribution of individual features to the predictions made by complex models. Here, we examine the results of LIME analysis for Logistic Regression, Support Vector Machine (SVM), K-nearest neighbors (KNN), Random Forest, Decision Tree, and Gradient Boosting classifiers, focusing on their impact on predicting risk levels.

4.6.1 Logistic Regression LIME Analysis

For the Logistic Regression model shown in Figure 20, the LIME analysis reveals that the feature group ≤ -0.51 has the most substantial positive contribution to the high-risk prediction, with a value of 0.43. This suggests that when the group feature is less than or equal to -0.51, it significantly increases the probability of a high-risk classification. Other essential features include Q23 ≤ -0.66 and Q20 ≤ -0.57 , contributing 0.11 and 0.07, respectively, to the high-risk

prediction. The remaining features, such as $Q22 \leq -0.91$ and $\text{sum_qchat} \leq -0.65$, also contribute positively, although to a lesser extent.

LIME Analysis for Logistic Regression:

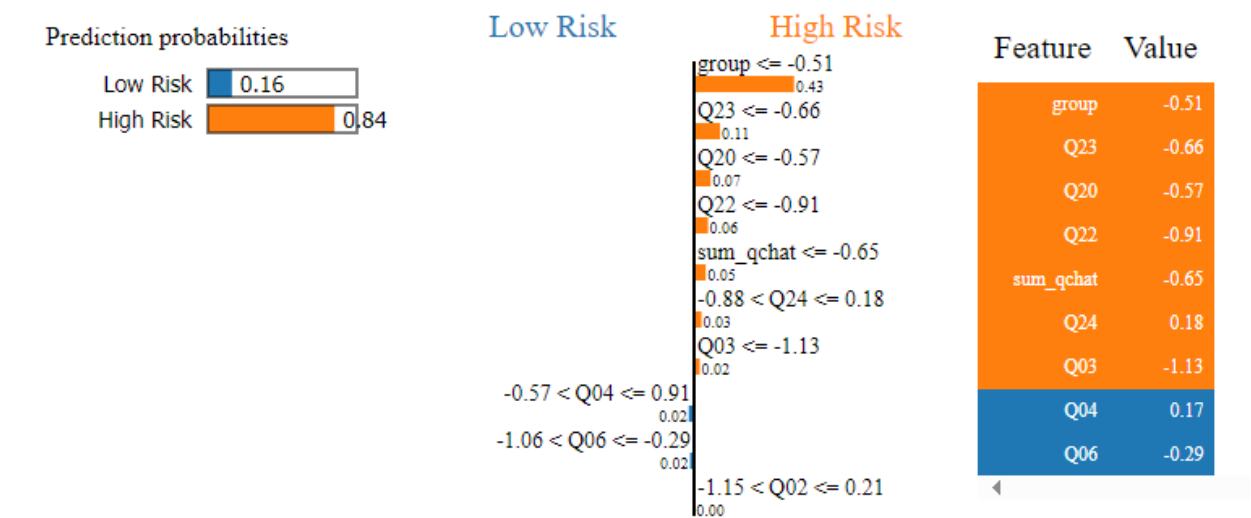


Figure 20: LIME Analysis for Logistic Regression Model: Feature Contributions to High-Risk Prediction

4.6.2 Support Vector Machine (SVM) LIME Analysis

In the case of the SVM model shown in Figure 21, the feature $\text{group} \leq -0.51$ again emerges as the most influential contributing 0.48 towards predicting high risk. This is followed by $Q20 \leq -0.57$ and $Q22 \leq -0.91$, contributing 0.13 and 0.12, respectively. The feature $\text{sum_qchat} \leq -0.65$ also has a notable impact, with a contribution of 0.07. Unlike the Logistic Regression model, SVM assigns a relatively lower contribution to some features such as $-0.57 < Q04 \leq 0.91$ and $-1.06 < Q06 \leq -0.29$.

LIME Analysis for SVM:

Prediction probabilities

Low Risk 0.00
High Risk 1.00

Low Risk

High Risk

group <= -0.51 0.48
Q20 <= -0.57 0.13
Q22 <= -0.91 0.12
Q23 <= -0.66 0.07
sum_qchat <= -0.65 0.07
-0.88 < Q24 <= 0.18 0.05
-0.57 < Q04 <= 0.91 0.03
-1.06 < Q06 <= -0.29 0.03
Q03 <= -1.13 0.01
-1.15 < Q02 <= 0.21 0.01

Feature Value

group	-0.51
Q20	-0.57
Q22	-0.91
Q23	-0.66
sum_qchat	-0.65
Q24	0.18
Q04	0.17
Q06	-0.29
Q03	-1.13

Figure 21: LIME Analysis for Support Vector Machine Model: Feature Contributions to High-Risk Prediction

4.6.3 k-Nearest Neighbors (KNN) LIME Analysis

For the KNN model shown in Figure 22, the analysis highlights that group <= -0.51 has the highest contribution to the high-risk prediction, with a value of 0.70. This reflects a strong association between this feature condition and high-risk outcomes. Other significant features include Q23 <= -0.66 and Q20 <= -0.57, contributing 0.08 and 0.06, respectively. The contributions of features such as Q22 <= -0.91 and sum_qchat <= -0.65 are also notable, although they are comparatively lower.

LIME Analysis for KNN:

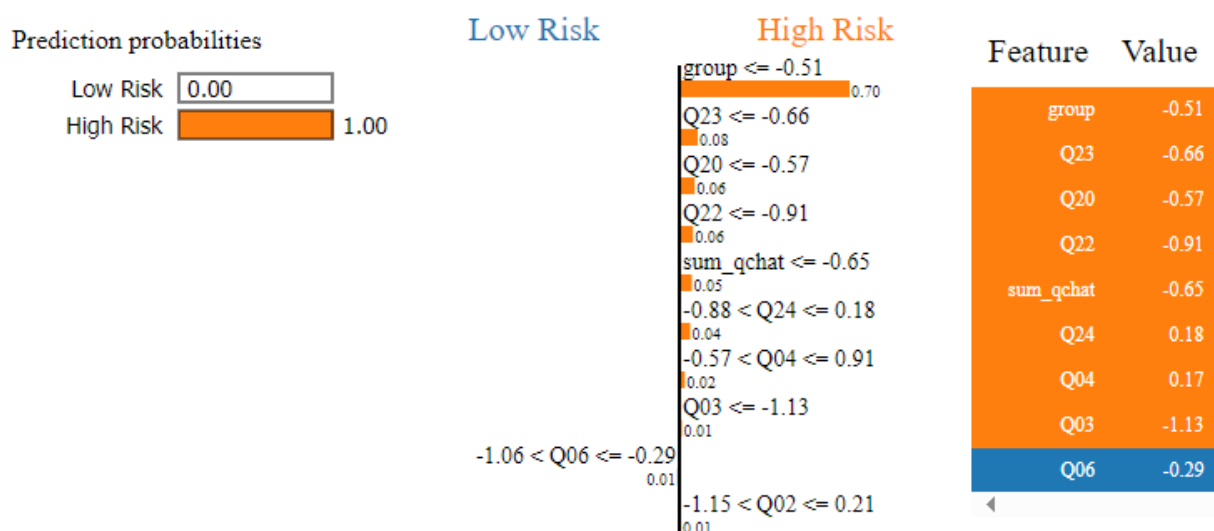


Figure 22: LIME Analysis for K-Nearest Neighbors Model: Feature Contributions to High-Risk Prediction

4.6.4 Random Forest LIME Analysis

In the Random Forest model shown in Figure 23, group <= -0.51 continues to be a major contributor with a value of 0.35 towards high-risk predictions. Features like Q23 <= -0.66 and Q20 <= -0.57 contribute 0.12 and 0.10, respectively. The feature -0.57 < Q04 <= 0.91 stands out with a contribution of 0.07, reflecting its importance in determining risk. Other features, such as sum_qchat <= -0.65 and Q24, also contribute, though to a lesser extent.

LIME Analysis for Random Forest:

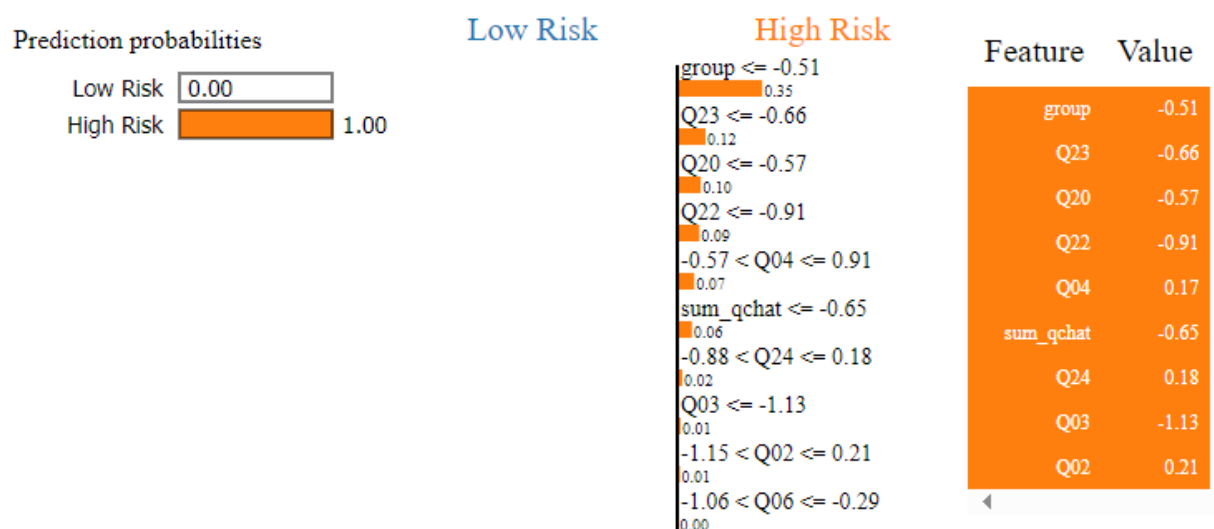


Figure 23: LIME Analysis for Random Forest Model: Feature Contributions to High-Risk Prediction

4.6.5 Decision Tree LIME Analysis

The Decision Tree model's LIME analysis shown in Figure 24 indicates that $\text{group} \leq -0.51$ is the most influential feature with a contribution of 0.42 to the high-risk prediction. $\text{Q23} \leq -0.66$ and $\text{Q20} \leq -0.57$ follow, with contributions of 0.17 and 0.13, respectively. The feature $\text{Q03} \leq -1.13$ also shows a small negative contribution of -0.04, suggesting that it slightly decreases the likelihood of a high-risk prediction. Other features such as $-0.88 < \text{Q24} \leq 0.18$ and $-1.06 < \text{Q06} \leq -0.29$ contribute marginally.

LIME Analysis for Decision Tree:

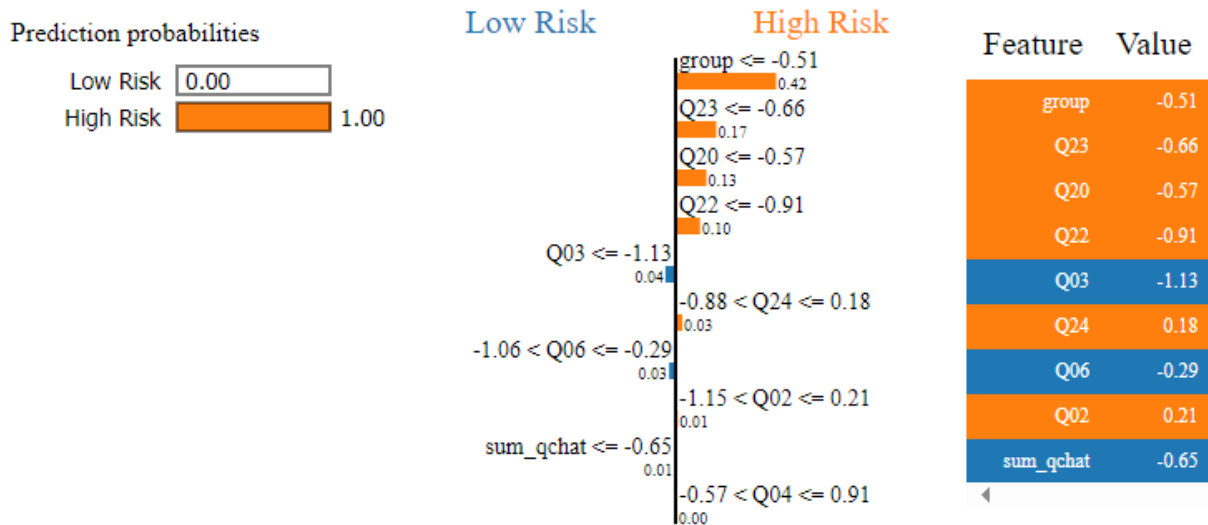


Figure 24: LIME Analysis for Decision Tree Model: Feature Contributions to High-Risk Prediction

4.6.6 Gradient Boosting LIME Analysis

For the Gradient Boosting model shown in Figure 25, $\text{group} \leq -0.51$ remains a significant predictor of high risk with a contribution of 0.38. Features like $\text{Q22} \leq -0.91$, $\text{Q20} \leq -0.57$, and $\text{Q23} \leq -0.66$ also contribute notably, with values of 0.13, 0.12, and 0.12, respectively. The analysis highlights the importance of $-0.57 < \text{Q04} \leq 0.91$ and $-0.88 < \text{Q24} \leq 0.18$, with contributions of 0.08 and 0.05. The features $\text{sum_qchat} \leq -0.65$ and $\text{Q03} \leq -1.13$ show smaller contributions but still play a role in the prediction.

LIME Analysis for Gradient Boosting:

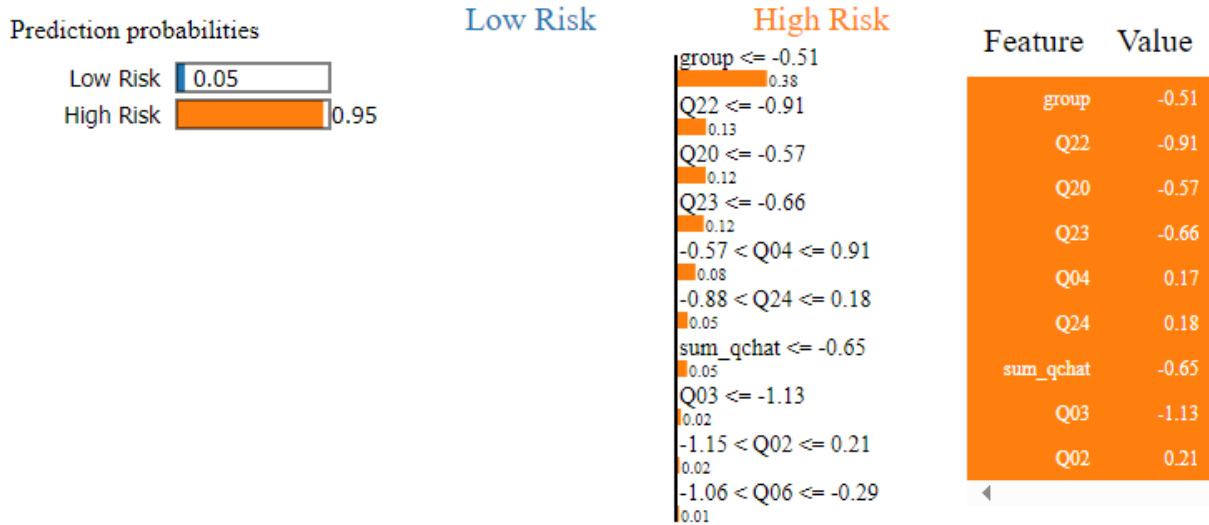


Figure 25: LIME Analysis for Gradient Boosting Model: Feature Contributions to High-Risk Prediction

4.7 Discussions

This study conducted a detailed evaluation of several machine learning models—Logistic Regression, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Decision Tree, Gradient Boosting, and Stacking Classifier—for detecting Autism Spectrum Disorder (ASD) in toddlers using the Q-CHAT dataset. The focus was on model performance, feature importance, and interpretability, acknowledging the study’s limitations.

4.7.1 Baseline Model Performance with Complete Feature Set

Initially, the models were assessed using the complete feature set from the Q-CHAT dataset. SVM demonstrated exceptional performance with an accuracy of 96.8%, a perfect recall of 1.0, and the highest ROC AUC score of 0.996, indicating its superior capability to distinguish between autistic and non-autistic cases. Random Forest closely followed with a 96.3% accuracy and strong performance across all metrics, including an ROC AUC of 0.990. Gradient Boosting, though slightly less accurate at 93.9%, showed strong recall and an ROC AUC of 0.977. KNN, while achieving perfect recall, had a lower precision and overall accuracy, with an ROC AUC of 0.963. Decision Tree and Logistic Regression had comparatively lower scores, with Decision Tree showing moderate performance and Logistic Regression being the least accurate among the

models. This initial evaluation set a benchmark for comparing advanced model configurations and feature reduction techniques.

4.7.2 Performance After Feature Selection

After the Recursive Feature Elimination (RFE), the models showed improved performance. SVM emerged as the top performer with an impressive accuracy of 99.2%, perfect recall, and an outstanding F1 score of 0.992. Random Forest and Gradient Boosting also exhibited strong results, with accuracies of 96.3% and 93.7%, respectively, and balanced precision and recall. KNN displayed a high recall rate but had lower precision and accuracy compared to the top models. Improvement in Logistic Regression and Decision Tree models was seen after feature selection but remained less competitive. RFE implementation enhanced model performance, validating the efficacy of the refined feature set.

4.7.3 Cross-Validation Performance Analysis

Cross-validation results confirmed the stability and reliability of the models after feature selection. SVM continued to lead with an average accuracy of 96.0%, high precision, perfect recall, and an average ROC AUC score of 99.4%. Random Forest and Gradient Boosting also demonstrated strong performance with high average accuracies and balanced metrics. KNN, Logistic Regression, and Decision Tree showed slightly lower accuracies but still provided solid performance in detecting autism risk. Overall, feature selection contributed to more robust and reliable results across models.

4.7.4 Stacking Classifier Performance

The stacking classifier, which integrates various base models with a Logistic Regression meta-classifier, achieved exceptional results with an accuracy of 98.9% and a perfect ROC AUC of 1.000. Cross-validation further confirmed its effectiveness, with an average accuracy of 97.0%, high precision, and recall. This model's performance highlights its ability to leverage the strengths of multiple base models, providing robust and reliable predictions.

4.7.5 Feature Importance Analysis

The feature importance analysis revealed that Q24, Q23, group, and Q20 features were consistently important across different models. Logistic Regression and SVM highlighted Q24 and Q23, while

tree-based models like Random Forest, Gradient Boosting, and Decision Tree also emphasized these features. This consistency emphasizes the significant role these features play in model predictions, offering a solid foundation for further development and application in ASD detection.

4.7.6 Local Interpretable Model-agnostic Explanations (LIME) Analysis

LIME analysis provided insights into how individual features contribute to model predictions. For Logistic Regression and SVM, the feature "group <= -0.51" was a major contributor to high-risk predictions, while KNN and Random Forest also highlighted this feature's importance. Decision Tree and Gradient Boosting showed similar trends, reinforcing the significance of "group <= -0.51" and other features like Q23 and Q20. LIME's ability to elucidate feature impacts enhances the interpretability of complex models, making their predictions more transparent.

4.8 Limitations of Study

The study relies on the Quantitative Checklist for Autism in Toddlers (Q-CHAT) dataset, which does not fully represent the diverse population of toddlers across different demographics and geographical regions. This limitation could affect the generalizability of the findings to broader populations. While the study employs Local Interpretable Model-agnostic Explanations (LIME) to enhance model interpretability, the complexity of some machine learning models, such as ensemble methods, can still pose challenges in fully understanding the decision-making process, particularly for clinicians without a technical background. The feature selection process, although rigorous, may introduce biases if certain features are overemphasized or underrepresented in the dataset. This could potentially skew the model's predictions and impact its reliability in real-world applications. The application of machine learning in clinical settings raises ethical concerns, particularly regarding data privacy and the potential for algorithmic bias. These issues need to be addressed to ensure the responsible use of AI in healthcare.

4.9 Response to Research Questions (RQ)

RQ1. How do various machine learning models perform in detecting Autism Spectrum Disorder (ASD) in toddlers using the Q-CHAT dataset?

Answer: The study assessed various machine learning models—including Logistic Regression, Random Forest, SVM, K-Nearest Neighbors, Decision Tree, Gradient Boosting, and a Stacking

Classifier—using the Q-CHAT dataset. The Stacking Classifier achieved the highest performance with 98.9% accuracy and a perfect ROC AUC of 1.000. SVM followed closely with 96.8% accuracy and a ROC AUC of 0.996. Random Forest and Gradient Boosting also performed well, with accuracies of 96.3% and 93.9%, respectively, highlighting the effectiveness of SVM and the Stacking Classifier in detecting ASD in toddlers.

RQ2. What is the impact of feature selection using Recursive Feature Elimination (RFE) on the performance of machine learning models for ASD detection?

Answer: Utilizing Recursive Feature Elimination (RFE) greatly improved model performance. Post-RFE, SVM reached an accuracy of 99.2% and an F1-score of 0.992, showcasing nearly perfect classification. Random Forest and Gradient Boosting also benefited, with accuracies of 96.3% and 93.7%, respectively. RFE enhanced model precision and reduced overfitting by focusing on the most relevant features, thus improving generalization.

RQ3. What are the most important features identified by machine learning models in predicting ASD risk using the Q-CHAT dataset?

Answer: Analysis of feature importance revealed that features such as Q24, Q23, group, and Q20 were consistently important across multiple models. Both Logistic Regression and SVM highlighted Q24 and Q23, while tree-based models also identified these features as critical. This consistent recognition underscores their significant role in predicting ASD risk.

RQ4. How does Local Interpretable Model-agnostic Explanations (LIME) contribute to the interpretability of machine learning models for ASD detection?

Answer: LIME enhanced the interpretability of machine learning models by clarifying the impact of individual features on predictions. For instance, in the Logistic Regression model, the feature "group <= -0.51" was notably influential in predicting high-risk cases. Similarly, in the SVM model, features like "group <= -0.51" and "Q20 <= -0.57" were crucial. LIME's insights into feature contributions make complex models more transparent and comprehensible for clinicians.

4.10 Implications of Study for Clinical Application

The study enhances clinical ASD diagnosis by using advanced machine learning models, improving detection accuracy with techniques like the Stacking Classifier and SVM. Feature

selection through Recursive Feature Elimination (RFE) boosts model performance by focusing on key predictors such as Q24, Q23, and Q20. Local Interpretable Model-agnostic Explanations (LIME) provide transparency, helping clinicians understand and trust the model's predictions. Overall, these advancements lead to more precise and comprehensible ASD diagnoses, improving early detection and intervention.

For practical applications, these models could be integrated into routine pediatric screenings and electronic health record (EHR) systems, enabling real-time alerts for clinicians when a high risk of ASD is detected. This integration would require careful planning, including collaboration with healthcare IT departments to ensure compatibility with existing systems. Scalability is also crucial, as the models must perform well across diverse clinical environments, including resource-limited settings like rural clinics. Cloud-based solutions could support real-time data processing and model application in such settings. Additionally, clinician training is essential to address potential resistance and ensure they are familiar with the models' strengths and limitations. This comprehensive approach aims to improve early detection and intervention for ASD, making these advanced tools accessible and effective across various healthcare settings.

4.11 Comparison with Other Studies

This study advances autism spectrum disorder (ASD) detection by achieving exceptional performance with the Stacking Classifier and SVM, showing accuracy and ROC AUC scores surpassing previous research. For instance, the Stacking Classifier's 98.9% accuracy and perfect ROC AUC outperform the high accuracy achieved in studies like Safahi et al. (2024) and Shrivastava et al. (2024). The study's use of Recursive Feature Elimination (RFE) also led to significant improvements, achieving 99.2% accuracy with SVM, a notable enhancement over methods reported by Abbas et al. (2018).

Feature importance analysis confirmed key predictors such as Q24, Q23, and Q20, aligning with findings from Duda et al. (2016) but showcasing their reliability in a new dataset. Additionally, the application of Local Interpretable Model-agnostic Explanations (LIME) in this study provides valuable insights into model decision-making, adding transparency to ASD detection processes. This comprehensive approach to accuracy, feature selection, and interpretability sets this study apart from previous research, offering significant improvements in ASD diagnostic tools.

5. Chapter Five: Conclusion, Recommendation, and Future Works

5.1 Conclusion

This study has made significant strides in the detection of Autism Spectrum Disorder (ASD) in toddlers through machine-learning methodologies. By evaluating various models including Logistic Regression, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Gradient Boosting, and a Stacking Classifier using the Q-CHAT dataset, the research highlights the superior performance of the Stacking Classifier and SVM. The Stacking Classifier achieved a remarkable accuracy of 98.9% and a perfect ROC AUC of 1.000, while SVM showed impressive accuracy at 96.8% with a ROC AUC of 0.996.

The study also demonstrated the effectiveness of Recursive Feature Elimination (RFE) in enhancing model performance, with SVM achieving a near-perfect accuracy of 99.2% post-feature selection. This indicates that RFE significantly refines models by focusing on relevant features, thus improving overall predictive accuracy. Feature importance analysis identified Q24, Q23, and Q20 as critical predictors of ASD risk, providing valuable insights into key indicators.

Furthermore, the use of Local Interpretable Model-agnostic Explanations (LIME) has enhanced the interpretability of machine learning models, making the decision-making process more transparent and actionable for clinicians. This study underscores the potential of advanced machine learning techniques to improve early ASD detection, offering a robust foundation for future diagnostic tools.

5.2 Recommendation

To enhance autism detection and diagnosis, it is recommended that advanced machine learning models, particularly the Stacking Classifier and SVM, be adopted due to their superior performance demonstrated in this study. Implementing Recursive Feature Elimination (RFE) should be prioritized to refine model accuracy by focusing on the most relevant features, thereby reducing overfitting and enhancing generalization. It is also crucial to emphasize the key features identified, such as Q24, Q23, and Q20, to improve the reliability of ASD risk assessments. Additionally, expanding the use of Local Interpretable Model-agnostic Explanations (LIME) will ensure that models are not only accurate but also interpretable, fostering greater trust and understanding among clinicians.

5.3 Future Works

Future research should focus on collecting and analyzing data from a more diverse range of populations to improve the generalizability of the models. This could involve collaborating with international research institutions to access varied datasets. Advanced interpretability techniques such as SHapley Additive exPlanations (SHAP), should be developed and integrated to provide deeper insights into model predictions and enhance clinician trust in AI-driven diagnostics. Real-world clinical trials should be conducted to test the efficacy and practicality of the developed models in live clinical environments. Exploring the integration of machine learning models with other diagnostic tools, such as genetic testing or neuroimaging, could provide a more comprehensive approach to ASD detection and improve diagnostic accuracy.

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